Bremen Smart Governance: Complete Technical Analysis

2024 Ortsteil statistical results; separate 2020 income context.

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Shared technical evidence, not final university report sections. Primary reporting year 2024; historical changes remain blocked. All figures and tables are reused from saved validated outputs.

1. Dataset Summary

Primary reporting year: 2024. Geographic level: Ortsteil. No imputation.

1. Dataset Summary

Dataset counts

Field	Value
Primary reporting year	2024
Workbook Ortsteile	88
Complete-case Ortsteile	79
Excluded Ortsteile	9
Baseline indicators	3
Imputed values	0

1. Dataset Summary

Statistical source observations, missing values and reporting years

name	records	ok values	missing values	invalid or suppressed values	duplicate key rows	unknown geography rows	years
csv_district_statistics_long.csv	15147	14924	223	0	0	146	2010-2016 (6 reporting years)
excel_ortsteilatlas_long.csv	316064	291709	24355	0	0	0	1969-2025 (50 reporting years)

1. Dataset Summary

Source reconstruction audit; JSON records are metadata

check	source	passed	records	missing
raw reconstruction equals existing processed output	excel_ortsteilatlas_long.csv	True	316064	24355.000
raw reconstruction equals existing processed output	csv_district_statistics_long.csv	True	15147	223.000
JSON metadata records readable separately	bremen_data.json	True	1515	Unavailable

1. Dataset Summary

Source inventory; catalogue years are title metadata, not observation years

dataset name	source type	years	processed records
ST111.csv	CSV district snapshot	2010-2016 (6 reporting years)	596
ST121.csv	CSV district snapshot	2010-2016 (6 reporting years)	1350
ST123.csv	CSV district snapshot	2010-2016 (6 reporting years)	745
ST124.csv	CSV district snapshot	2010-2016 (6 reporting years)	750
ST125.csv	CSV district snapshot	2010-2016 (6 reporting years)	441
ST126.csv	CSV district snapshot	2010-2016 (6 reporting years)	144
ST127.csv	CSV district snapshot	2010-2016 (6 reporting years)	143
ST131.csv	CSV district snapshot	2010-2016 (6 reporting years)	735
ST132.csv	CSV district snapshot	2010-2016 (6 reporting years)	1176
ST133.csv	CSV district snapshot	2010-2016 (6 reporting years)	740
ST134.csv	CSV district snapshot	2010-2016 (6 reporting years)	600
ST135.csv	CSV district snapshot	2010-2016 (6 reporting years)	148
ST136.csv	CSV district snapshot	2010-2015 (5 reporting years)	144
ST137.csv	CSV district snapshot	2010-2016 (6 reporting years)	888
ST138.csv	CSV district snapshot	2010-2016 (6 reporting years)	900
ST141.csv	CSV district snapshot	2010-2016 (6 reporting years)	146

1. Dataset Summary

Source inventory; catalogue years are title metadata, not observation years

dataset name	source type	years	processed records
ST142.csv	CSV district snapshot	2010-2016 (6 reporting years)	745
ST143.csv	CSV district snapshot	2010-2016 (6 reporting years)	1168
ST144.csv	CSV district snapshot	2010-2016 (6 reporting years)	894
ST151.csv	CSV district snapshot	2010-2016 (6 reporting years)	894
ST152.csv	CSV district snapshot	2010-2016 (6 reporting years)	900
ST153.csv	CSV district snapshot	2010-2016 (6 reporting years)	900
Datentabelle.xls::Gebiet und Fläche	Excel worksheet	1988-2024 (26 reporting years)	9184
Datentabelle.xls::Bevölkerung (Absolute Werte)	Excel worksheet	1974-2024 (19 reporting years)	19152
Datentabelle.xls::Altersstruktur I	Excel worksheet	1974-2024 (19 reporting years)	19152
Datentabelle.xls::Altersstruktur II	Excel worksheet	1974-2024 (19 reporting years)	13888
Datentabelle.xls::Familienstand und Wohndauer	Excel worksheet	1980-2024 (25 reporting years)	10304
Datentabelle.xls::Ausländische Bevölkerung	Excel worksheet	1974-2024 (27 reporting years)	4368
Datentabelle.xls::Migrationshintergrund	Excel worksheet	2005-2024 (20 reporting years)	24640
Datentabelle.xls::Bevölkerungsbewegung	Excel worksheet	1979-2024 (10 reporting years)	1120
Datentabelle.xls::Geburten und Sterbefälle	Excel worksheet	1974-2024 (19 reporting years)	10640
Datentabelle.xls::Wanderungen	Excel worksheet	2000-2024 (16 reporting years)	10752

1. Dataset Summary

Source inventory; catalogue years are title metadata, not observation years

dataset name	source type	years	processed records
Datentabelle.xls::Wanderungssalden	Excel worksheet	2000-2024 (16 reporting years)	10752
Datentabelle.xls::Privathaushalte	Excel worksheet	2005-2024 (20 reporting years)	13440
Datentabelle.xls::Kirche	Excel worksheet	1994-2021 (17 reporting years)	1904
Datentabelle.xls::Bildung	Excel worksheet	2012-2024 (13 reporting years)	23744
Datentabelle.xls::Erwerbstätigkeit, Arbeitslosig	Excel worksheet	2009-2024 (16 reporting years)	15232
Datentabelle.xls::Betriebe und Beschäftigte	Excel worksheet	2009-2021 (12 reporting years)	14560
Datentabelle.xls::Wohnungswesen I	Excel worksheet	1988-2024 (18 reporting years)	17472
Datentabelle.xls::Wohnungswesen II (Ergebnisse d	Excel worksheet	2011-2022 (2 reporting years)	672
Datentabelle.xls::Heizen und Energieträger (Erge	Excel worksheet	2011-2022 (2 reporting years)	1568
Datentabelle.xls::Bautätigkeit	Excel worksheet	1983-2023 (21 reporting years)	4704
Datentabelle.xls::Verkehr	Excel worksheet	2005-2024 (20 reporting years)	7616
Datentabelle.xls::Gesundheit	Excel worksheet	1983-2025 (21 reporting years)	8624
Datentabelle.xls::Sozialleistungen	Excel worksheet	2005-2024 (20 reporting years)	22400
Datentabelle.xls::Lohn- und Einkommensteuer	Excel worksheet	2001-2020 (12 reporting years)	6720
Datentabelle.xls::Insolvenzen	Excel worksheet	2022-2024 (3 reporting years)	336
Datentabelle.xls::Bundestagswahlen	Excel worksheet	1969-2025 (16 reporting years)	23072

1. Dataset Summary

Source inventory; catalogue years are title metadata, not observation years

dataset name	source type	years	processed records
Datentabelle.xls::Bürgerschaftswahlen	Excel worksheet	1971-2023 (14 reporting years)	20048
bremen_data.json	JSON open-data metadata catalogue	1991-2027 (31 reporting years)	1515

1. Dataset Summary

Excluded Ortsteile, 2024

geo code	geo name	exclusion reason
122	Industriehäfen	unemployment:missing;employment:missing;benefit_receipt:missing
124	Neustädter Hafen	unemployment:missing;employment:missing;benefit_receipt:missing
125	Hohentorshafen	unemployment:missing;employment:missing;benefit_receipt:missing
271	Strom	unemployment:missing;employment:missing
411	Blockland	unemployment:missing;employment:missing
424	In den Hufen	unemployment:missing;employment:missing
436	Hohweg	unemployment:missing;employment:missing
444	In den Wischen	unemployment:missing;employment:missing;benefit_receipt:missing
512	Werderland	unemployment:missing;employment:missing

1. Dataset Summary

Validation metadata

Field	Status
CSV format	CP1252; semicolon; decimal comma
Excel format	BIFF XLS; xlrd; numeric cells
Missing data	Retained; complete-case scoring
Aggregate geographic levels	Excluded from scoring
Name alias	Überseestadt / Ueberseestadt: code 437
Historical change calculations	Not produced; boundary and definition stability unverified
Within-year reference dates	Unverified

2. Indicators and Coverage

Baseline: unemployment, employment, benefit_receipt. Coverage proportions are fractions.

2. Indicators and Coverage

Indicators: original names, units, available years and scoring direction

feature	original german name	english name	unit	available years	role	disadvantage direction
unemployment	Anteil der Arbeitslosen an den sozialversicherungspflichtig Beschäftigten zuzüglich Arbeitslose (%)	Unemployment quotient (%)	%	2009; 2010; 2011; 2012; 2013; 2014; 2015; 2016; 2017; 2018; 2019; 2020; 2021; 2022; 2023; 2024	primary	higher
employment	Anteil der sozialversicherungspflichtig Beschäftigten an der Bevölkerung im Alter von 15 bis unter 65 Jahren (%)	Social-insurance employment (%)	%	2009; 2010; 2011; 2012; 2013; 2014; 2015; 2016; 2017; 2018; 2019; 2020; 2021; 2022; 2023; 2024	primary	lower
benefit_receipt	Anteil der Leistungsberechtigten (Grundsicherung für Arbeitsuchende, SGB II) an der Bevölkerung unter 65 Jahren (%)	SGB II receipt, under 65 (%)	%	2005; 2006; 2007; 2008; 2009; 2010; 2011; 2012; 2013; 2014; 2015; 2016; 2017; 2018; 2019; 2020; 2021; 2022; 2023; 2024	primary	higher
child_benefits	Anteil der Leistungsberechtigten (Grundsicherung für Arbeitsuchende, SGB II) an der Bevölkerung unter 15 Jahren (%)	SGB II receipt, under 15 (%)	%	2005; 2006; 2007; 2008; 2009; 2010; 2011; 2012; 2013; 2014; 2015; 2016; 2017; 2018; 2019; 2020; 2021; 2022; 2023; 2024	sensitivity	higher
income_median	Gesamtbetrag der Einkünfte je Steuerpflichtigen (1000 EUR): Median	Median income per taxpayer (1000 EUR)	1000 EUR	2001; 2004; 2007; 2010; 2013; 2014; 2015; 2016; 2017; 2018; 2019; 2020	historical_context	lower
single_parents	Anteil der Alleinerziehenden an den Privathaushalten mit Hauptwohnsitz mit Kind(ern) (%)	Single-parent households (%)	%	2005; 2006; 2007; 2008; 2009; 2010; 2011; 2012; 2013; 2014; 2015; 2016; 2017; 2018; 2019; 2020; 2021; 2022; 2023; 2024	context_only	not_scored

2. Indicators and Coverage

Measurement definitions

feature	measurement definition
unemployment	Registered unemployed / (social-insurance employees + registered unemployed).
employment	Social-insurance employees at residence / population aged 15 to under 65; excludes self-employment and civil servants.
benefit_receipt	SGB II entitled recipients / population under 65.
child_benefits	SGB II entitled recipients under 15 / population under 15.
income_median	Median total income per taxpayer, thousand EUR.
single_parents	Single-parent households / main-residence households with children.

2. Indicators and Coverage

Indicator sources and latest coverage

feature	source file	source sheet	latest year	latest covered	total ortsteile	latest coverage proportion	latest missing rate
unemployment	Datentabelle.xls	Erwerbstätigkeit, Arbeitslosig	2024	79	88	0.898	0.102
employment	Datentabelle.xls	Erwerbstätigkeit, Arbeitslosig	2024	79	88	0.898	0.102
benefit_receipt	Datentabelle.xls	Sozialleistungen	2024	84	88	0.955	0.045
child_benefits	Datentabelle.xls	Sozialleistungen	2024	84	88	0.955	0.045
income_median	Datentabelle.xls	Lohn- und Einkommensteuer	2020	84	88	0.955	0.045
single_parents	Datentabelle.xls	Privathaushalte	2024	83	88	0.943	0.057

2. Indicators and Coverage

Coverage by indicator and reporting year

feature	year	total ortsteile	observed	coverage proportion	missing	missing rate
unemployment	2009	88	84	0.955	4	0.045
unemployment	2010	88	0	0.000	88	1.000
unemployment	2011	88	84	0.955	4	0.045
unemployment	2012	88	84	0.955	4	0.045
unemployment	2013	88	84	0.955	4	0.045
unemployment	2014	88	85	0.966	3	0.034
unemployment	2015	88	85	0.966	3	0.034
unemployment	2016	88	85	0.966	3	0.034
unemployment	2017	88	85	0.966	3	0.034
unemployment	2018	88	79	0.898	9	0.102
unemployment	2019	88	79	0.898	9	0.102
unemployment	2020	88	79	0.898	9	0.102
unemployment	2021	88	79	0.898	9	0.102
unemployment	2022	88	79	0.898	9	0.102
unemployment	2023	88	79	0.898	9	0.102
unemployment	2024	88	79	0.898	9	0.102

2. Indicators and Coverage

Coverage by indicator and reporting year

feature	year	total ortsteile	observed	coverage proportion	missing	missing rate
employment	2009	88	84	0.955	4	0.045
employment	2010	88	84	0.955	4	0.045
employment	2011	88	84	0.955	4	0.045
employment	2012	88	84	0.955	4	0.045
employment	2013	88	84	0.955	4	0.045
employment	2014	88	85	0.966	3	0.034
employment	2015	88	85	0.966	3	0.034
employment	2016	88	85	0.966	3	0.034
employment	2017	88	85	0.966	3	0.034
employment	2018	88	79	0.898	9	0.102
employment	2019	88	79	0.898	9	0.102
employment	2020	88	79	0.898	9	0.102
employment	2021	88	79	0.898	9	0.102
employment	2022	88	79	0.898	9	0.102
employment	2023	88	79	0.898	9	0.102
employment	2024	88	79	0.898	9	0.102

2. Indicators and Coverage

Coverage by indicator and reporting year

feature	year	total ortsteile	observed	coverage proportion	missing	missing rate
benefit_receipt	2005	88	84	0.955	4	0.045
benefit_receipt	2006	88	84	0.955	4	0.045
benefit_receipt	2007	88	84	0.955	4	0.045
benefit_receipt	2008	88	84	0.955	4	0.045
benefit_receipt	2009	88	84	0.955	4	0.045
benefit_receipt	2010	88	84	0.955	4	0.045
benefit_receipt	2011	88	84	0.955	4	0.045
benefit_receipt	2012	88	84	0.955	4	0.045
benefit_receipt	2013	88	84	0.955	4	0.045
benefit_receipt	2014	88	85	0.966	3	0.034
benefit_receipt	2015	88	85	0.966	3	0.034
benefit_receipt	2016	88	85	0.966	3	0.034
benefit_receipt	2017	88	85	0.966	3	0.034
benefit_receipt	2018	88	84	0.955	4	0.045
benefit_receipt	2019	88	84	0.955	4	0.045
benefit_receipt	2020	88	84	0.955	4	0.045

2. Indicators and Coverage

Coverage by indicator and reporting year

feature	year	total ortsteile	observed	coverage proportion	missing	missing rate
benefit_receipt	2021	88	84	0.955	4	0.045
benefit_receipt	2022	88	84	0.955	4	0.045
benefit_receipt	2023	88	84	0.955	4	0.045
benefit_receipt	2024	88	84	0.955	4	0.045
child_benefits	2005	88	84	0.955	4	0.045
child_benefits	2006	88	84	0.955	4	0.045
child_benefits	2007	88	84	0.955	4	0.045
child_benefits	2008	88	84	0.955	4	0.045
child_benefits	2009	88	82	0.932	6	0.068
child_benefits	2010	88	81	0.920	7	0.080
child_benefits	2011	88	82	0.932	6	0.068
child_benefits	2012	88	82	0.932	6	0.068
child_benefits	2013	88	83	0.943	5	0.057
child_benefits	2014	88	85	0.966	3	0.034
child_benefits	2015	88	85	0.966	3	0.034
child_benefits	2016	88	85	0.966	3	0.034

2. Indicators and Coverage

Coverage by indicator and reporting year

feature	year	total ortsteile	observed	coverage proportion	missing	missing rate
child_benefits	2017	88	85	0.966	3	0.034
child_benefits	2018	88	84	0.955	4	0.045
child_benefits	2019	88	83	0.943	5	0.057
child_benefits	2020	88	84	0.955	4	0.045
child_benefits	2021	88	83	0.943	5	0.057
child_benefits	2022	88	84	0.955	4	0.045
child_benefits	2023	88	83	0.943	5	0.057
child_benefits	2024	88	84	0.955	4	0.045
income_median	2001	88	84	0.955	4	0.045
income_median	2004	88	85	0.966	3	0.034
income_median	2007	88	83	0.943	5	0.057
income_median	2010	88	84	0.955	4	0.045
income_median	2013	88	84	0.955	4	0.045
income_median	2014	88	84	0.955	4	0.045
income_median	2015	88	84	0.955	4	0.045
income_median	2016	88	84	0.955	4	0.045

2. Indicators and Coverage

Coverage by indicator and reporting year

feature	year	total ortsteile	observed	coverage proportion	missing	missing rate
income_median	2017	88	84	0.955	4	0.045
income_median	2018	88	84	0.955	4	0.045
income_median	2019	88	84	0.955	4	0.045
income_median	2020	88	84	0.955	4	0.045
single_parents	2005	88	83	0.943	5	0.057
single_parents	2006	88	83	0.943	5	0.057
single_parents	2007	88	83	0.943	5	0.057
single_parents	2008	88	83	0.943	5	0.057
single_parents	2009	88	84	0.955	4	0.045
single_parents	2010	88	84	0.955	4	0.045
single_parents	2011	88	84	0.955	4	0.045
single_parents	2012	88	84	0.955	4	0.045
single_parents	2013	88	84	0.955	4	0.045
single_parents	2014	88	85	0.966	3	0.034
single_parents	2015	88	84	0.955	4	0.045
single_parents	2016	88	84	0.955	4	0.045

2. Indicators and Coverage

Coverage by indicator and reporting year

feature	year	total ortsteile	observed	coverage proportion	missing	missing rate
single_parents	2017	88	84	0.955	4	0.045
single_parents	2018	88	84	0.955	4	0.045
single_parents	2019	88	83	0.943	5	0.057
single_parents	2020	88	83	0.943	5	0.057
single_parents	2021	88	83	0.943	5	0.057
single_parents	2022	88	83	0.943	5	0.057
single_parents	2023	88	83	0.943	5	0.057
single_parents	2024	88	83	0.943	5	0.057

3. Descriptive Statistics

2024; unweighted neighborhood statistics; sample standard deviation (ddof=1); rates in percent.

3. Descriptive Statistics

Complete-case statistics, 2024; percent

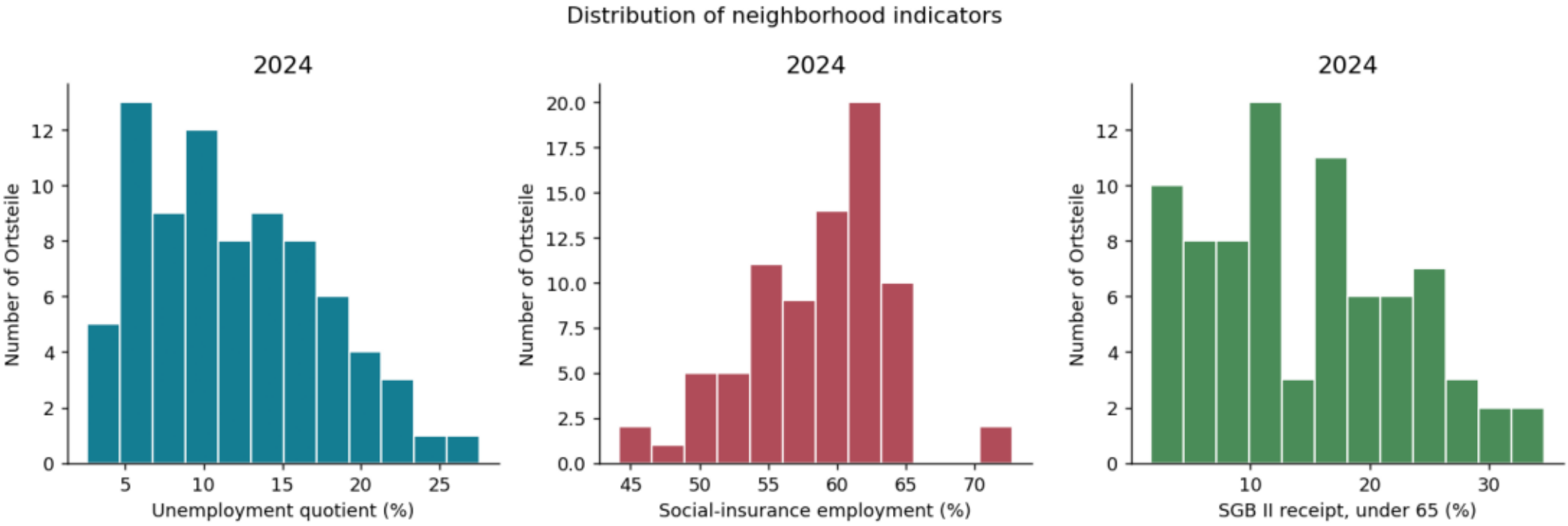
feature	count	mean	std	min	25%	median	75%	max
unemployment	79.000	11.862	5.554	2.590	7.409	10.976	16.170	27.541
employment	79.000	58.630	5.313	44.159	55.644	59.404	62.209	72.714
benefit_receipt	79.000	14.844	8.392	1.702	8.304	13.232	20.532	34.622

3. Descriptive Statistics

Available-case statistics, 2024; percent

feature	count	mean	std	min	25%	median	75%	max
unemployment	79.000	11.862	5.554	2.590	7.409	10.976	16.170	27.541
employment	79.000	58.630	5.313	44.159	55.644	59.404	62.209	72.714
benefit_receipt	84.000	14.329	8.571	0.990	6.573	12.469	20.104	34.622

3. Descriptive Statistics



Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Complete cases; unweighted neighborhoods; descriptive, not causal.

2024 indicator distributions; percent; complete cases.

3. Descriptive Statistics

Spearman indicator correlations; unitless

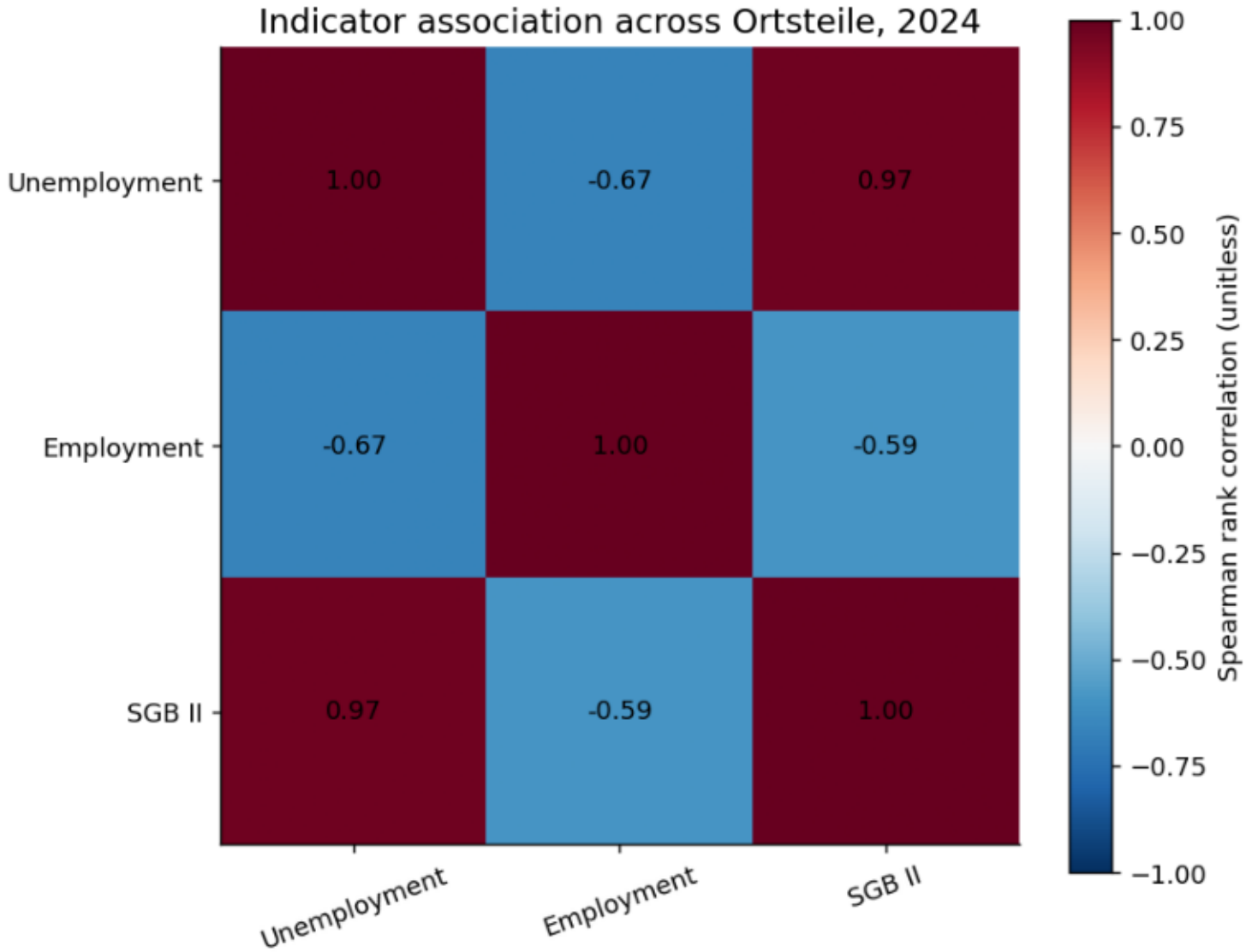
feature	unemployment	employment	benefit receipt
unemployment	1.000	-0.667	0.974
employment	-0.667	1.000	-0.588
benefit_receipt	0.974	-0.588	1.000

3. Descriptive Statistics

Pearson indicator correlations; unitless

feature	unemployment	employment	benefit receipt
unemployment	1.000	-0.679	0.971
employment	-0.679	1.000	-0.602
benefit_receipt	0.971	-0.602	1.000

3. Descriptive Statistics



Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Correlation does not imply causation; indicators overlap conceptually.

2024 Spearman correlations; unitless.

3. Descriptive Statistics

1.5-IQR flags; observations retained

geo code	geo name	year	feature	value	lower fence	upper fence
252	Rablinghausen	2024	employment	72.401	45.797	72.056
261	Seehausen	2024	employment	72.714	45.797	72.056
522	Grohn	2024	employment	44.186	45.797	72.056
525	Fähr-Lobbendorf	2024	employment	44.159	45.797	72.056

4. Neighborhood Comparisons

2024 complete-case neighborhoods; 2020 income context shown separately.

4. Neighborhood Comparisons

All included Ortsteile, 2024; indicator values in percent

geo code	geo name	year	unemployment	employment	benefit receipt
111	Altstadt	2024	9.913	63.620	13.232
112	Bahnhofsvorstadt	2024	20.261	54.538	26.383
113	Ostertor	2024	9.740	56.038	9.368
211	Alte Neustadt	2024	10.775	63.061	11.807
212	Hohentor	2024	14.162	58.835	16.998
213	Neustadt	2024	8.540	59.404	9.146
214	Südervorstadt	2024	10.857	61.732	11.600
215	Gartenstadt Süd	2024	11.075	62.897	16.518
216	Buntentor	2024	9.159	62.254	9.200
217	Neuenland	2024	21.419	58.484	23.539
218	Huckelriede	2024	14.204	49.470	18.416
231	Habenhausen	2024	3.963	63.226	2.410
232	Arsten	2024	8.187	63.586	10.476
233	Kattenturm	2024	17.219	54.852	23.642
234	Kattenesch	2024	8.179	63.174	10.220
241	Mittelshuchting	2024	16.763	55.472	25.788

4. Neighborhood Comparisons

All included Ortsteile, 2024; indicator values in percent

geo code	geo name	year	unemployment	employment	benefit receipt
242	Sodenmatt	2024	18.321	55.169	26.838
243	Kirchhuchting	2024	18.968	51.741	25.884
244	Grolland	2024	5.155	63.804	4.378
251	Woltmershausen	2024	13.723	59.911	17.353
252	Rablinghausen	2024	9.073	72.401	10.984
261	Seehausen	2024	2.590	72.714	2.854
311	Steintor	2024	8.126	58.080	7.624
312	Fesenfeld	2024	7.475	60.342	6.671
313	Peterswerder	2024	6.632	61.253	5.580
314	Hulsberg	2024	10.044	61.826	10.911
321	Neu-Schwachhausen	2024	6.196	62.748	6.278
322	Bürgerpark	2024	4.928	57.174	3.157
323	Barkhof	2024	5.301	62.662	3.311
324	Riensberg	2024	6.056	61.623	6.170
325	Radio Bremen	2024	6.601	59.367	5.751
326	Schwachhausen	2024	4.181	60.236	3.352

4. Neighborhood Comparisons

All included Ortsteile, 2024; indicator values in percent

geo code	geo name	year	unemployment	employment	benefit receipt
327	Gete	2024	5.686	60.326	3.865
331	Gartenstadt Vahr	2024	9.177	62.747	10.936
332	Neue Vahr Nord	2024	18.006	56.709	26.273
334	Neue Vahr Südwest	2024	12.428	63.210	19.786
335	Neue Vahr Südost	2024	16.707	59.078	24.470
341	Horn	2024	4.569	60.347	3.551
342	Lehe	2024	6.676	54.140	6.251
343	Lehesterdeich	2024	7.342	62.475	8.880
351	Borgfeld	2024	3.446	57.103	1.702
361	Oberneuland	2024	5.466	56.202	4.471
371	Ellener Feld	2024	8.104	61.930	11.245
372	Ellenerbrok-Schevemoor	2024	11.845	60.042	17.284
373	Tenever	2024	19.776	51.290	30.255
374	Osterholz	2024	4.734	65.107	3.611
375	Blockdiek	2024	15.494	58.099	20.961
381	Sebaldsbrück	2024	10.976	60.789	13.639

4. Neighborhood Comparisons

All included Ortsteile, 2024; indicator values in percent

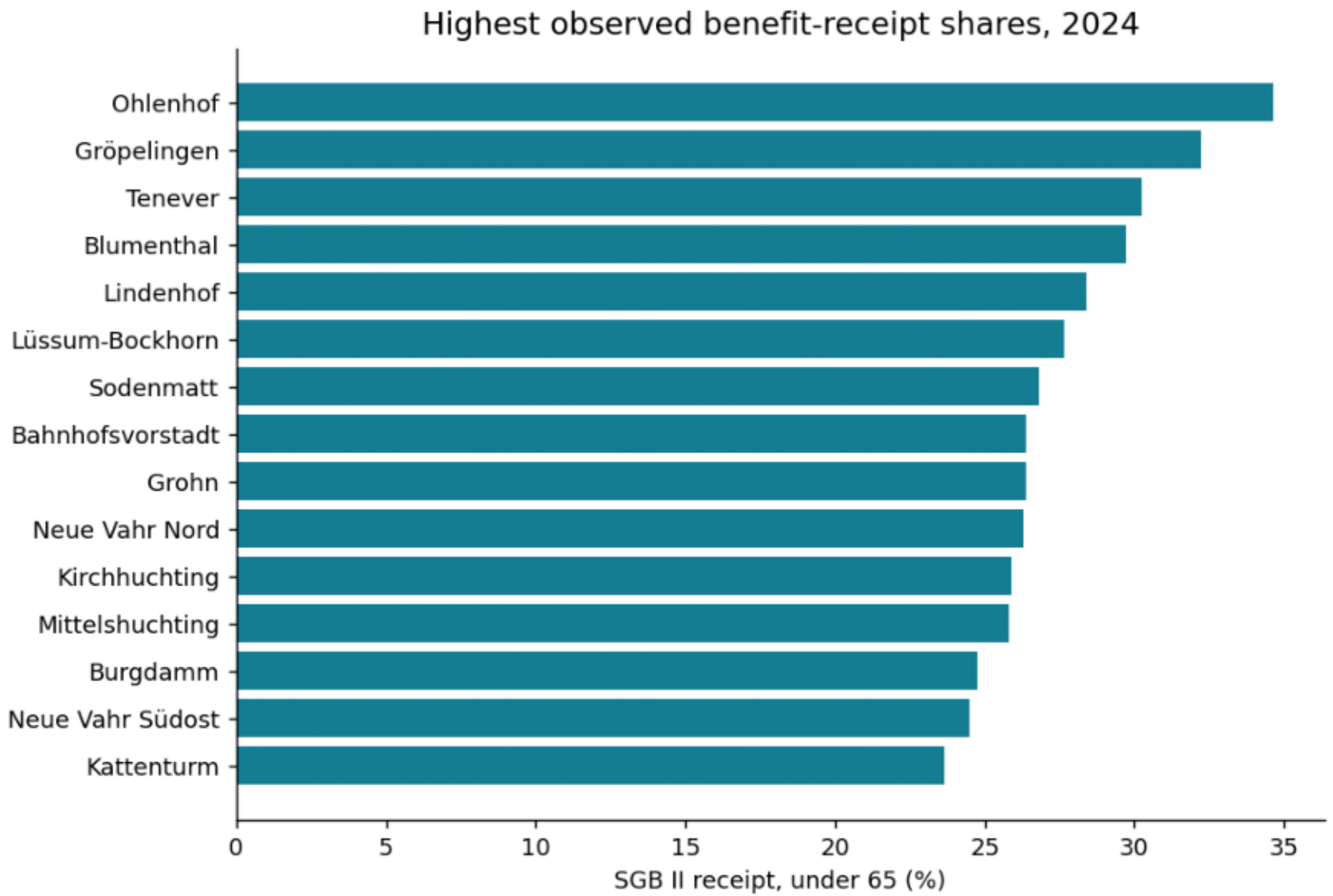
geo code	geo name	year	unemployment	employment	benefit receipt
382	Hastedt	2024	11.541	61.562	12.419
383	Hemelingen	2024	16.227	56.767	23.563
384	Arbergen	2024	6.643	63.656	9.925
385	Mahndorf	2024	9.290	64.645	12.379
421	Regensburger Straße	2024	9.175	61.227	11.662
422	Findorff-Bürgerweide	2024	8.470	64.017	10.686
423	Weidedamm	2024	7.217	62.163	8.661
431	Utbremen	2024	11.504	61.832	20.103
432	Steffensweg	2024	12.259	60.838	16.919
433	Westend	2024	13.601	55.952	17.802
434	Walle	2024	14.991	58.781	18.028
435	Osterfeuerberg	2024	16.113	55.816	17.878
437	Ueberseestadt	2024	14.129	47.633	19.422
441	Lindenhof	2024	22.208	51.809	28.411
442	Gröpelingen	2024	25.264	50.042	32.208
443	Ohlenhof	2024	27.541	49.060	34.622

4. Neighborhood Comparisons

All included Ortsteile, 2024; indicator values in percent

geo code	geo name	year	unemployment	employment	benefit receipt
445	Oslebshausen	2024	16.602	53.512	22.259
511	Burg-Grambke	2024	13.613	62.097	18.256
513	Burgdamm	2024	17.269	55.396	24.745
514	Lesum	2024	11.052	56.480	15.057
515	St. Magnus	2024	9.229	51.704	6.176
521	Vegesack	2024	16.240	54.589	21.585
522	Grohn	2024	20.443	44.186	26.370
523	Schönebeck	2024	10.061	58.998	12.518
524	Aumund-Hammersbeck	2024	15.716	55.972	19.866
525	Fähr-Lobbendorf	2024	17.411	44.159	17.669
531	Blumenthal	2024	21.983	51.119	29.737
532	Rönnebeck	2024	13.940	58.120	17.436
533	Lüssum-Bockhorn	2024	19.354	52.392	27.639
534	Farge	2024	13.203	60.960	17.698
535	Rekum	2024	6.571	65.023	7.947

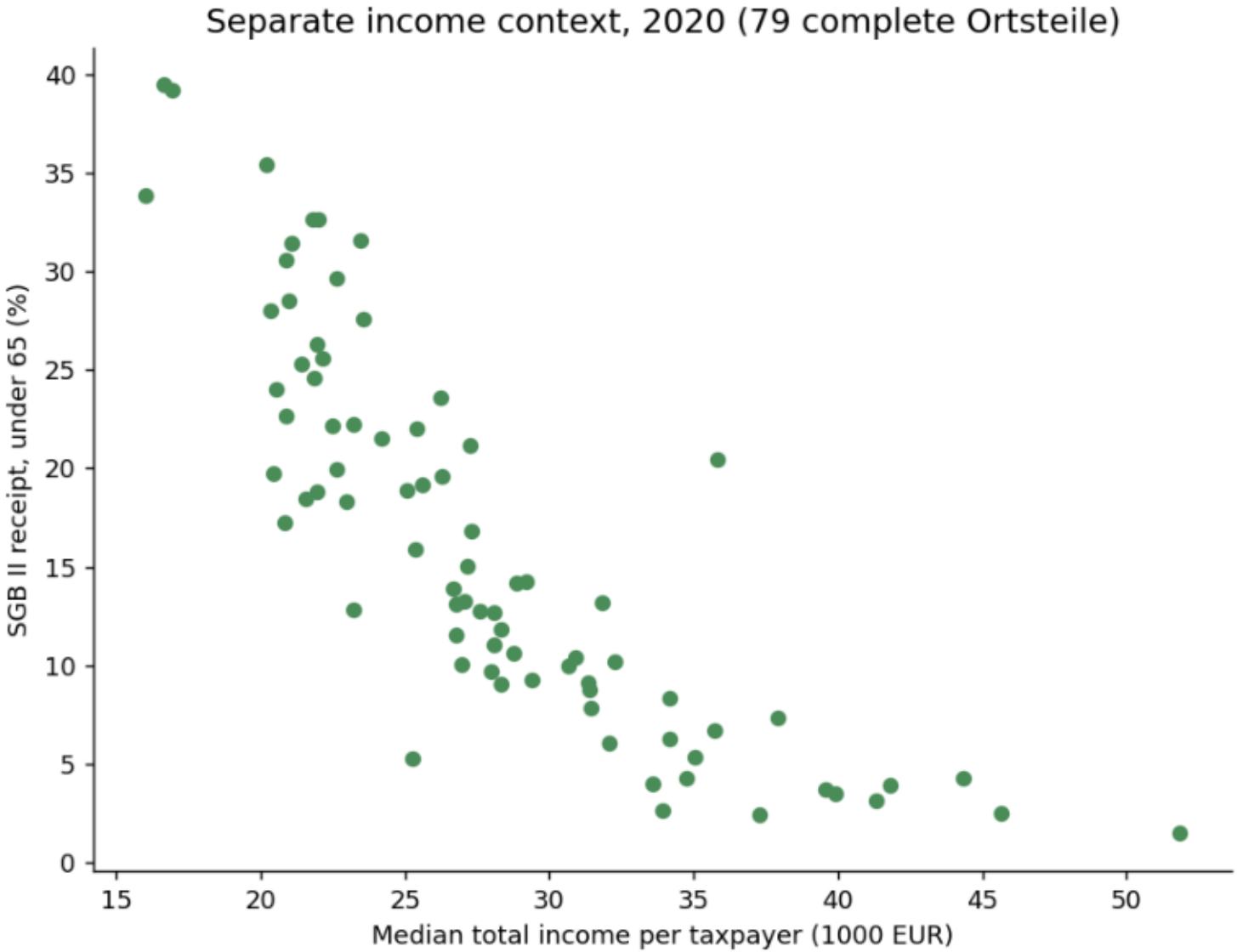
4. Neighborhood Comparisons



Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Complete cases; unweighted neighborhoods; descriptive, not causal.

2024 under-65 SGB II receipt; percent; highest 15 observed shares.

4. Neighborhood Comparisons



Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
2020 context only; not joined to 2024 scores. Taxpayer income is not household disposable income.

2020 only; median income per taxpayer in 1000 EUR; SGB II receipt in percent.

5. Composite Index

Directed min-max normalization; equal weights (1/3 each); score 0-1; descending competition ranks.

5. Composite Index

Complete composite scores and ranks, 2024; score 0-1

geo code	geo name	year	composite score	rank
443	Ohlenhof	2024	0.943	1
442	Gröpelingen	2024	0.876	2
522	Grohn	2024	0.821	3
531	Blumenthal	2024	0.795	4
441	Lindenhof	2024	0.777	5
373	Tenever	2024	0.769	6
533	Lüssum-Bockhorn	2024	0.724	7
243	Kirchhuchting	2024	0.708	8
112	Bahnhofsvorstadt	2024	0.698	9
525	Fähr-Lobbendorf	2024	0.693	10
242	Sodenmatt	2024	0.669	11
332	Neue Vahr Nord	2024	0.642	12
217	Neuenland	2024	0.639	13
241	Mittelshuchting	2024	0.635	14
513	Burgdamm	2024	0.632	15
437	Ueberseestadt	2024	0.626	16

5. Composite Index

Complete composite scores and ranks, 2024; score 0-1

geo code	geo name	year	composite score	rank
233	Kattenturm	2024	0.626	17
445	Oslebshausen	2024	0.620	18
218	Huckelriede	2024	0.596	19
521	Vegesack	2024	0.595	20
383	Hemelingen	2024	0.590	21
335	Neue Vahr Südost	2024	0.578	22
524	Aumund-Hammersbeck	2024	0.555	23
435	Osterfeuerberg	2024	0.542	24
375	Blockdiek	2024	0.538	25
433	Westend	2024	0.506	26
434	Walle	2024	0.494	27
532	Rönnebeck	2024	0.481	28
212	Hohentor	2024	0.471	29
251	Woltmershausen	2024	0.457	30
534	Farge	2024	0.441	31
511	Burg-Grambke	2024	0.439	32

5. Composite Index

Complete composite scores and ranks, 2024; score 0-1

geo code	geo name	year	composite score	rank
514	Lesum	2024	0.438	33
431	Utbremen	2024	0.432	34
372	Ellenerbrok-Schevemoor	2024	0.429	35
334	Neue Vahr Südwest	2024	0.425	36
432	Steffensweg	2024	0.422	37
515	St. Magnus	2024	0.379	38
215	Gartenstadt Süd	2024	0.378	39
381	Sebaldsbrück	2024	0.372	40
523	Schönebeck	2024	0.369	41
113	Ostertor	2024	0.368	42
382	Hastedt	2024	0.358	43
214	Südervorstadt	2024	0.339	44
211	Alte Neustadt	2024	0.324	45
421	Regensburger Straße	2024	0.323	46
111	Altstadt	2024	0.321	47
314	Hulsberg	2024	0.320	48

5. Composite Index

Complete composite scores and ranks, 2024; score 0-1

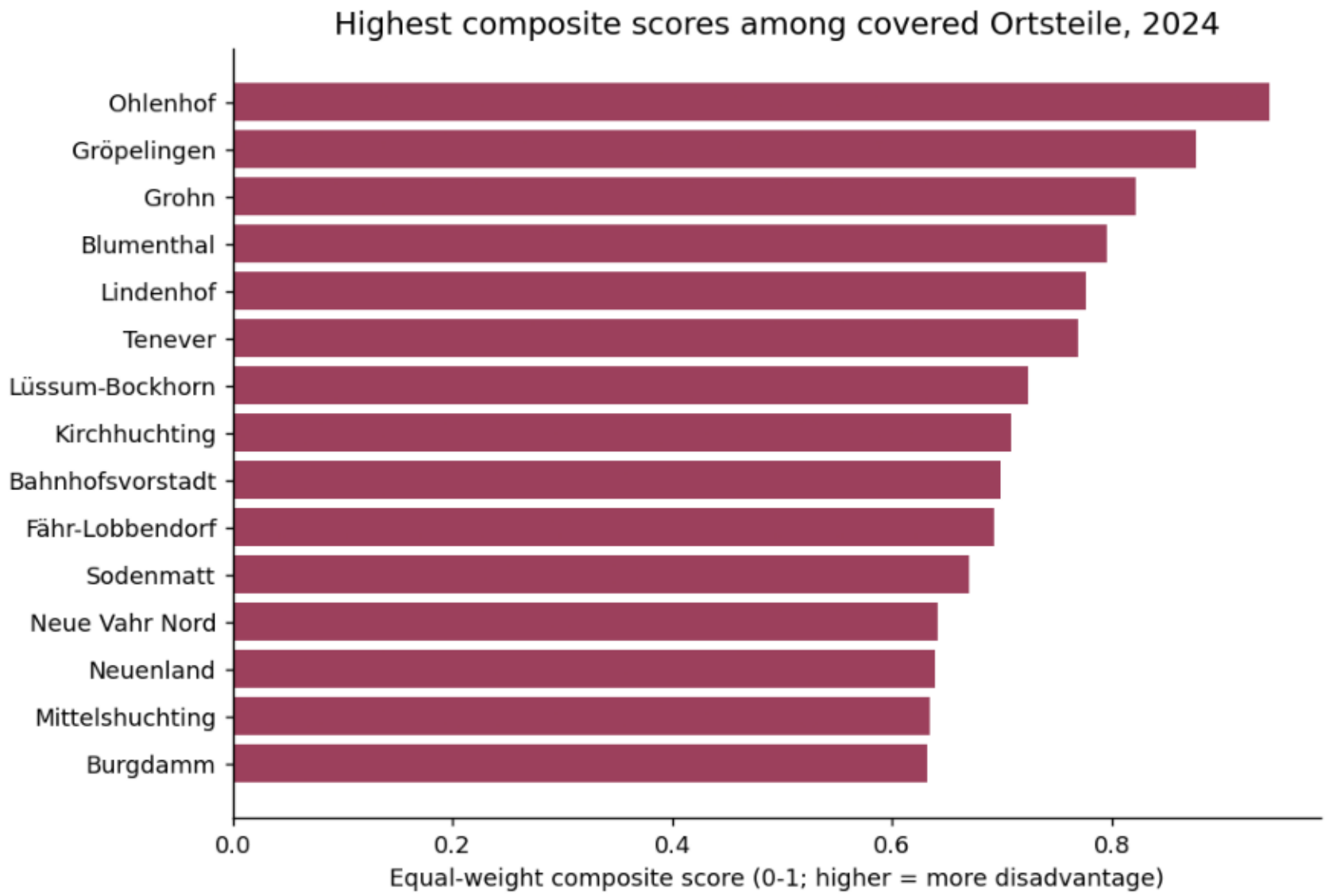
geo code	geo name	year	composite score	rank
342	Lehe	2024	0.317	49
213	Neustadt	2024	0.310	50
311	Steintor	2024	0.305	51
331	Gartenstadt Vahr	2024	0.298	52
371	Ellener Feld	2024	0.296	53
385	Mahndorf	2024	0.292	54
216	Buntentor	2024	0.286	55
234	Kattenesch	2024	0.272	56
422	Findorff-Bürgerweide	2024	0.271	57
232	Arsten	2024	0.270	58
312	Fesenfeld	2024	0.260	59
361	Oberneuland	2024	0.259	60
343	Lehesterdeich	2024	0.256	61
423	Weidedamm	2024	0.255	62
325	Radio Bremen	2024	0.250	63
384	Arbergen	2024	0.243	64

5. Composite Index

Complete composite scores and ranks, 2024; score 0-1

geo code	geo name	year	composite score	rank
322	Bürgerpark	2024	0.227	65
313	Peterswerder	2024	0.227	66
324	Riensberg	2024	0.221	67
321	Neu-Schwachhausen	2024	0.211	68
327	Gete	2024	0.208	69
535	Rekum	2024	0.206	70
351	Borgfeld	2024	0.194	71
341	Horn	2024	0.190	72
252	Rablinghausen	2024	0.184	73
326	Schwachhausen	2024	0.184	74
323	Barkhof	2024	0.170	75
244	Grolland	2024	0.165	76
374	Osterholz	2024	0.137	77
231	Habenhausen	2024	0.136	78
261	Seehausen	2024	0.012	79

5. Composite Index



Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Complete cases; unweighted neighborhoods; descriptive, not causal.

2024 composite scores; equal indicator weights; highest 15 scores.

5. Composite Index

Directed normalized indicators, 2024; 0-1

geo code	geo name	year	unemployment	employment	benefit receipt
111	Altstadt	2024	0.293	0.318	0.350
112	Bahnhofsvorstadt	2024	0.708	0.637	0.750
113	Ostertor	2024	0.287	0.584	0.233
211	Alte Neustadt	2024	0.328	0.338	0.307
212	Hohentor	2024	0.464	0.486	0.465
213	Neustadt	2024	0.238	0.466	0.226
214	Südervorstadt	2024	0.331	0.385	0.301
215	Gartenstadt Süd	2024	0.340	0.344	0.450
216	Buntentor	2024	0.263	0.366	0.228
217	Neuenland	2024	0.755	0.498	0.663
218	Huckelriede	2024	0.465	0.814	0.508
231	Habenhausen	2024	0.055	0.332	0.022
232	Arsten	2024	0.224	0.320	0.267
233	Kattenturm	2024	0.586	0.626	0.666
234	Kattenesch	2024	0.224	0.334	0.259
241	Mittelshuchting	2024	0.568	0.604	0.732

5. Composite Index

Directed normalized indicators, 2024; 0-1

geo code	geo name	year	unemployment	employment	benefit receipt
242	Sodenmatt	2024	0.630	0.614	0.764
243	Kirchhuchting	2024	0.656	0.734	0.735
244	Grolland	2024	0.103	0.312	0.081
251	Woltmershausen	2024	0.446	0.448	0.475
252	Rablinghausen	2024	0.260	0.011	0.282
261	Seehausen	2024	0.000	0.000	0.035
311	Steintor	2024	0.222	0.512	0.180
312	Fesenfeld	2024	0.196	0.433	0.151
313	Peterswerder	2024	0.162	0.401	0.118
314	Hulsberg	2024	0.299	0.381	0.280
321	Neu-Schwachhausen	2024	0.145	0.349	0.139
322	Bürgerpark	2024	0.094	0.544	0.044
323	Barkhof	2024	0.109	0.352	0.049
324	Riensberg	2024	0.139	0.388	0.136
325	Radio Bremen	2024	0.161	0.467	0.123
326	Schwachhausen	2024	0.064	0.437	0.050

5. Composite Index

Directed normalized indicators, 2024; 0-1

geo code	geo name	year	unemployment	employment	benefit receipt
327	Gete	2024	0.124	0.434	0.066
331	Gartenstadt Vahr	2024	0.264	0.349	0.281
332	Neue Vahr Nord	2024	0.618	0.560	0.746
334	Neue Vahr Südwest	2024	0.394	0.333	0.549
335	Neue Vahr Südost	2024	0.566	0.478	0.692
341	Horn	2024	0.079	0.433	0.056
342	Lehe	2024	0.164	0.650	0.138
343	Lehesterdeich	2024	0.190	0.359	0.218
351	Borgfeld	2024	0.034	0.547	0.000
361	Oberneuland	2024	0.115	0.578	0.084
371	Ellener Feld	2024	0.221	0.378	0.290
372	Ellenerbrok-Schevemoor	2024	0.371	0.444	0.473
373	Tenever	2024	0.689	0.750	0.867
374	Osterholz	2024	0.086	0.266	0.058
375	Blockdiek	2024	0.517	0.512	0.585
381	Sebaldsbrück	2024	0.336	0.418	0.363

5. Composite Index

Directed normalized indicators, 2024; 0-1

geo code	geo name	year	unemployment	employment	benefit receipt
382	Hastedt	2024	0.359	0.391	0.326
383	Hemelingen	2024	0.547	0.558	0.664
384	Arbergen	2024	0.162	0.317	0.250
385	Mahndorf	2024	0.269	0.283	0.324
421	Regensburger Straße	2024	0.264	0.402	0.303
422	Findorff-Bürgerweide	2024	0.236	0.305	0.273
423	Weidedamm	2024	0.185	0.369	0.211
431	Utbremen	2024	0.357	0.381	0.559
432	Steffensweg	2024	0.388	0.416	0.462
433	Westend	2024	0.441	0.587	0.489
434	Walle	2024	0.497	0.488	0.496
435	Osterfeuerberg	2024	0.542	0.592	0.491
437	Ueberseestadt	2024	0.462	0.878	0.538
441	Lindenhof	2024	0.786	0.732	0.811
442	Gröpelingen	2024	0.909	0.794	0.927
443	Ohlenhof	2024	1.000	0.828	1.000

5. Composite Index

Directed normalized indicators, 2024; 0-1

geo code	geo name	year	unemployment	employment	benefit receipt
445	Oslebshausen	2024	0.562	0.672	0.624
511	Burg-Grambke	2024	0.442	0.372	0.503
513	Burgdamm	2024	0.588	0.606	0.700
514	Lesum	2024	0.339	0.568	0.406
515	St. Magnus	2024	0.266	0.736	0.136
521	Vegesack	2024	0.547	0.635	0.604
522	Grohn	2024	0.716	0.999	0.749
523	Schönebeck	2024	0.299	0.480	0.329
524	Aumund-Hammersbeck	2024	0.526	0.586	0.552
525	Fähr-Lobbendorf	2024	0.594	1.000	0.485
531	Blumenthal	2024	0.777	0.756	0.852
532	Rönnebeck	2024	0.455	0.511	0.478
533	Lüssum-Bockhorn	2024	0.672	0.712	0.788
534	Farge	2024	0.425	0.412	0.486
535	Rekum	2024	0.160	0.269	0.190

6. TOPSIS

Column vector normalization; equal weights (1/3 each); Euclidean distance; descending closeness ranks.

6. TOPSIS

Complete TOPSIS closeness, ranks and normalized distances, 2024

geo code	geo name	year	topsis score	rank	distance to disadvantage ideal	distance to advantage ideal
443	Ohlenhof	2024	0.971	1	0.003	0.103
442	Gröpelingen	2024	0.911	2	0.009	0.095
531	Blumenthal	2024	0.810	3	0.020	0.084
441	Lindenhof	2024	0.796	4	0.021	0.082
373	Tenever	2024	0.767	5	0.025	0.081
522	Grohn	2024	0.738	6	0.027	0.077
533	Lüssum-Bockhorn	2024	0.726	7	0.029	0.076
112	Bahnhofsvorstadt	2024	0.726	8	0.028	0.075
217	Neuenland	2024	0.699	9	0.031	0.073
243	Kirchhuchting	2024	0.696	10	0.032	0.072
242	Sodenmatt	2024	0.691	11	0.032	0.072
332	Neue Vahr Nord	2024	0.676	12	0.034	0.071
241	Mittelshuchting	2024	0.645	13	0.037	0.068
513	Burgdamm	2024	0.642	14	0.037	0.067
233	Kattenturm	2024	0.626	15	0.039	0.065
335	Neue Vahr Südost	2024	0.623	16	0.039	0.065

6. TOPSIS

Complete TOPSIS closeness, ranks and normalized distances, 2024

geo code	geo name	year	topsis score	rank	distance to disadvantage ideal	distance to advantage ideal
383	Hemelingen	2024	0.603	17	0.041	0.063
445	Oslebshausen	2024	0.595	18	0.042	0.062
521	Vegesack	2024	0.577	19	0.044	0.060
525	Fähr-Lobbendorf	2024	0.551	20	0.047	0.058
375	Blockdiek	2024	0.550	21	0.047	0.057
524	Aumund-Hammersbeck	2024	0.541	22	0.048	0.056
435	Osterfeuerberg	2024	0.519	23	0.050	0.054
437	Ueberseestadt	2024	0.512	24	0.051	0.054
218	Huckelriede	2024	0.497	25	0.052	0.052
434	Walle	2024	0.496	26	0.052	0.051
511	Burg-Grambke	2024	0.470	27	0.055	0.049
433	Westend	2024	0.469	28	0.055	0.049
334	Neue Vahr Südwest	2024	0.469	29	0.056	0.049
532	Rönnebeck	2024	0.468	30	0.055	0.048

6. TOPSIS

Complete TOPSIS closeness, ranks and normalized distances, 2024

geo code	geo name	year	topsis score	rank	distance to disadvantage ideal	distance to advantage ideal
212	Hohentor	2024	0.465	31	0.055	0.048
251	Woltmershausen	2024	0.461	32	0.056	0.048
431	Utbremen	2024	0.459	33	0.057	0.048
534	Farge	2024	0.455	34	0.057	0.047
432	Steffensweg	2024	0.426	35	0.060	0.044
372	Ellenerbrok-Schevemoor	2024	0.424	36	0.060	0.044
215	Gartenstadt Süd	2024	0.396	37	0.063	0.041
514	Lesum	2024	0.380	38	0.064	0.040
381	Sebaldsbrück	2024	0.352	39	0.067	0.036
382	Hastedt	2024	0.344	40	0.068	0.036
111	Altstadt	2024	0.323	41	0.070	0.033
523	Schönebeck	2024	0.320	42	0.071	0.033
214	Südervorstadt	2024	0.318	43	0.071	0.033
211	Alte Neustadt	2024	0.318	44	0.071	0.033
385	Mahndorf	2024	0.297	45	0.073	0.031

6. TOPSIS

Complete TOPSIS closeness, ranks and normalized distances, 2024

geo code	geo name	year	topsis score	rank	distance to disadvantage ideal	distance to advantage ideal
314	Hulsberg	2024	0.292	46	0.073	0.030
421	Regensburger Straße	2024	0.288	47	0.074	0.030
331	Gartenstadt Vahr	2024	0.275	48	0.075	0.028
113	Ostertor	2024	0.274	49	0.076	0.029
252	Rablinghausen	2024	0.266	50	0.076	0.028
371	Ellener Feld	2024	0.262	51	0.077	0.027
422	Findorff-Bürgerweide	2024	0.257	52	0.077	0.027
216	Buntentor	2024	0.250	53	0.078	0.026
232	Arsten	2024	0.249	54	0.078	0.026
234	Kattenesch	2024	0.245	55	0.078	0.025
213	Neustadt	2024	0.242	56	0.079	0.025
515	St. Magnus	2024	0.236	57	0.082	0.025
311	Steintor	2024	0.216	58	0.082	0.023
384	Arbergen	2024	0.214	59	0.082	0.022

6. TOPSIS

Complete TOPSIS closeness, ranks and normalized distances, 2024

geo code	geo name	year	topsis score	rank	distance to disadvantage ideal	distance to advantage ideal
343	Lehesterdeich	2024	0.211	60	0.082	0.022
423	Weidedamm	2024	0.206	61	0.082	0.021
312	Fesenfeld	2024	0.186	62	0.085	0.019
342	Lehe	2024	0.183	63	0.087	0.019
535	Rekum	2024	0.179	64	0.085	0.019
325	Radio Bremen	2024	0.161	65	0.088	0.017
313	Peterswerder	2024	0.155	66	0.088	0.016
321	Neu-Schwachhausen	2024	0.152	67	0.088	0.016
324	Riensberg	2024	0.150	68	0.089	0.016
361	Oberneuland	2024	0.138	69	0.092	0.015
327	Gete	2024	0.121	70	0.093	0.013
322	Bürgerpark	2024	0.115	71	0.095	0.012
244	Grolland	2024	0.105	72	0.093	0.011
323	Barkhof	2024	0.101	73	0.095	0.011
341	Horn	2024	0.099	74	0.096	0.011

6. TOPSIS

Complete TOPSIS closeness, ranks and normalized distances, 2024

geo code	geo name	year	topsis score	rank	distance to disadvantage ideal	distance to advantage ideal
326	Schwachhausen	2024	0.093	75	0.097	0.010
351	Borgfeld	2024	0.093	76	0.100	0.010
374	Osterholz	2024	0.085	77	0.095	0.009
231	Habenhausen	2024	0.070	78	0.099	0.007
261	Seehausen	2024	0.024	79	0.102	0.003

6. TOPSIS

TOPSIS weights and ideal profiles

indicator	weight	ideal disadvantage	anti ideal
unemployment	0.333	0.079	0.007
employment	0.333	0.028	0.046
benefit_receipt	0.333	0.076	0.004

6. TOPSIS

TOPSIS weighted normalized matrix, 2024

geo code	geo name	unemployment	employment	benefit receipt
111	Altstadt	0.028	0.041	0.029
112	Bahnhofsvorstadt	0.058	0.035	0.058
113	Ostertor	0.028	0.036	0.021
211	Alte Neustadt	0.031	0.040	0.026
212	Hohentor	0.041	0.037	0.037
213	Neustadt	0.024	0.038	0.020
214	Südervorstadt	0.031	0.039	0.026
215	Gartenstadt Süd	0.032	0.040	0.036
216	Buntentor	0.026	0.040	0.020
217	Neuenland	0.061	0.037	0.052
218	Huckelriede	0.041	0.032	0.041
231	Habenhausen	0.011	0.040	0.005
232	Arsten	0.023	0.041	0.023
233	Kattenturm	0.049	0.035	0.052
234	Kattenesch	0.023	0.040	0.023
241	Mittelshuchting	0.048	0.035	0.057

6. TOPSIS

TOPSIS weighted normalized matrix, 2024

geo code	geo name	unemployment	employment	benefit receipt
242	Sodenmatt	0.053	0.035	0.059
243	Kirchhuchting	0.054	0.033	0.057
244	Grolland	0.015	0.041	0.010
251	Woltmershausen	0.039	0.038	0.038
252	Rablinghausen	0.026	0.046	0.024
261	Seehausen	0.007	0.046	0.006
311	Steintor	0.023	0.037	0.017
312	Fesenfeld	0.021	0.038	0.015
313	Peterswerder	0.019	0.039	0.012
314	Hulsberg	0.029	0.039	0.024
321	Neu-Schwachhausen	0.018	0.040	0.014
322	Bürgerpark	0.014	0.036	0.007
323	Barkhof	0.015	0.040	0.007
324	Riensberg	0.017	0.039	0.014
325	Radio Bremen	0.019	0.038	0.013
326	Schwachhausen	0.012	0.038	0.007

6. TOPSIS

TOPSIS weighted normalized matrix, 2024

geo code	geo name	unemployment	employment	benefit receipt
327	Gete	0.016	0.038	0.009
331	Gartenstadt Vahr	0.026	0.040	0.024
332	Neue Vahr Nord	0.052	0.036	0.058
334	Neue Vahr Südwest	0.036	0.040	0.044
335	Neue Vahr Südost	0.048	0.038	0.054
341	Horn	0.013	0.038	0.008
342	Lehe	0.019	0.034	0.014
343	Lehesterdeich	0.021	0.040	0.020
351	Borgfeld	0.010	0.036	0.004
361	Oberneuland	0.016	0.036	0.010
371	Ellener Feld	0.023	0.039	0.025
372	Ellenerbrok-Schevemoor	0.034	0.038	0.038
373	Tenever	0.057	0.033	0.067
374	Osterholz	0.014	0.041	0.008
375	Blockdiek	0.044	0.037	0.046
381	Sebaldsbrück	0.031	0.039	0.030

6. TOPSIS

TOPSIS weighted normalized matrix, 2024

geo code	geo name	unemployment	employment	benefit receipt
382	Hastedt	0.033	0.039	0.027
383	Hemelingen	0.047	0.036	0.052
384	Arbergen	0.019	0.041	0.022
385	Mahndorf	0.027	0.041	0.027
421	Regensburger Straße	0.026	0.039	0.026
422	Findorff-Bürgerweide	0.024	0.041	0.024
423	Weidedamm	0.021	0.040	0.019
431	Utbremen	0.033	0.039	0.044
432	Steffensweg	0.035	0.039	0.037
433	Westend	0.039	0.036	0.039
434	Walle	0.043	0.037	0.040
435	Osterfeuerberg	0.046	0.036	0.039
437	Ueberseestadt	0.041	0.030	0.043
441	Lindenhof	0.064	0.033	0.063
442	Gröpelingen	0.072	0.032	0.071
443	Ohlenhof	0.079	0.031	0.076

6. TOPSIS

TOPSIS weighted normalized matrix, 2024

geo code	geo name	unemployment	employment	benefit receipt
445	Oslebshausen	0.048	0.034	0.049
511	Burg-Grambke	0.039	0.040	0.040
513	Burgdamm	0.050	0.035	0.055
514	Lesum	0.032	0.036	0.033
515	St. Magnus	0.026	0.033	0.014
521	Vegesack	0.047	0.035	0.048
522	Grohn	0.059	0.028	0.058
523	Schönebeck	0.029	0.038	0.028
524	Aumund-Hammersbeck	0.045	0.036	0.044
525	Fähr-Lobbendorf	0.050	0.028	0.039
531	Blumenthal	0.063	0.033	0.066
532	Rönnebeck	0.040	0.037	0.038
533	Lüssum-Bockhorn	0.055	0.033	0.061
534	Farge	0.038	0.039	0.039
535	Rekum	0.019	0.041	0.018

6. TOPSIS



2024 TOPSIS and composite ranks.

7. K-means

StandardScaler (ddof=0); KMeans; k-means++; n_init=30; random_state=42; nominal profile labels.

7. K-means

Cluster-count diagnostics

k	silhouette	inertia	calinski harabasz	davies bouldin	minimum cluster size
2	0.528	90.098	125.546	0.666	28
3	0.384	64.469	101.696	0.802	20
4	0.347	51.977	88.992	0.911	13
5	0.331	41.133	88.093	0.964	9
6	0.355	33.827	87.691	0.833	4
7	0.367	27.896	89.948	0.805	4
8	0.389	21.809	100.080	0.743	2

7. K-means

Cluster sizes and mean percentages, 2024

cluster	n	unemployment mean	employment mean	benefit receipt mean
Profile A	51	8.449	61.404	9.823
Profile B	28	18.077	53.579	23.989

7. K-means

All cluster assignments, 2024; labels are nominal

geo code	geo name	year	cluster
111	Altstadt	2024	Profile A
113	Ostertor	2024	Profile A
211	Alte Neustadt	2024	Profile A
212	Hohentor	2024	Profile A
213	Neustadt	2024	Profile A
214	Südervorstadt	2024	Profile A
215	Gartenstadt Süd	2024	Profile A
216	Buntentor	2024	Profile A
231	Habenhausen	2024	Profile A
232	Arsten	2024	Profile A
234	Kattenesch	2024	Profile A
244	Grolland	2024	Profile A
251	Woltmershausen	2024	Profile A
252	Rablinghausen	2024	Profile A
261	Seehausen	2024	Profile A
311	Steintor	2024	Profile A

7. K-means

All cluster assignments, 2024; labels are nominal

geo code	geo name	year	cluster
312	Fesenfeld	2024	Profile A
313	Peterswerder	2024	Profile A
314	Hulsberg	2024	Profile A
321	Neu-Schwachhausen	2024	Profile A
322	Bürgerpark	2024	Profile A
323	Barkhof	2024	Profile A
324	Riensberg	2024	Profile A
325	Radio Bremen	2024	Profile A
326	Schwachhausen	2024	Profile A
327	Gete	2024	Profile A
331	Gartenstadt Vahr	2024	Profile A
334	Neue Vahr Südwest	2024	Profile A
341	Horn	2024	Profile A
342	Lehe	2024	Profile A
343	Lehesterdeich	2024	Profile A
351	Borgfeld	2024	Profile A

7. K-means

All cluster assignments, 2024; labels are nominal

geo code	geo name	year	cluster
361	Oberneuland	2024	Profile A
371	Ellener Feld	2024	Profile A
372	Ellenerbrok-Schevemoor	2024	Profile A
374	Osterholz	2024	Profile A
381	Sebaldsbrück	2024	Profile A
382	Hastedt	2024	Profile A
384	Arbergen	2024	Profile A
385	Mahndorf	2024	Profile A
421	Regensburger Straße	2024	Profile A
422	Findorff-Bürgerweide	2024	Profile A
423	Weidedamm	2024	Profile A
431	Utbremen	2024	Profile A
432	Steffensweg	2024	Profile A
511	Burg-Grambke	2024	Profile A
514	Lesum	2024	Profile A
515	St. Magnus	2024	Profile A

7. K-means

All cluster assignments, 2024; labels are nominal

geo code	geo name	year	cluster
523	Schönebeck	2024	Profile A
534	Farge	2024	Profile A
535	Rekum	2024	Profile A
112	Bahnhofsvorstadt	2024	Profile B
217	Neuenland	2024	Profile B
218	Huckelriede	2024	Profile B
233	Kattenturm	2024	Profile B
241	Mittelshuchting	2024	Profile B
242	Sodenmatt	2024	Profile B
243	Kirchhuchting	2024	Profile B
332	Neue Vahr Nord	2024	Profile B
335	Neue Vahr Südost	2024	Profile B
373	Tenever	2024	Profile B
375	Blockdiek	2024	Profile B
383	Hemelingen	2024	Profile B
433	Westend	2024	Profile B

7. K-means

All cluster assignments, 2024; labels are nominal

geo code	geo name	year	cluster
434	Walle	2024	Profile B
435	Osterfeuerberg	2024	Profile B
437	Ueberseestadt	2024	Profile B
441	Lindenhof	2024	Profile B
442	Gröpelingen	2024	Profile B
443	Ohlenhof	2024	Profile B
445	Oslebshausen	2024	Profile B
513	Burgdamm	2024	Profile B
521	Vegesack	2024	Profile B
522	Grohn	2024	Profile B
524	Aumund-Hammersbeck	2024	Profile B
525	Fähr-Lobbendorf	2024	Profile B
531	Blumenthal	2024	Profile B
532	Rönnebeck	2024	Profile B
533	Lüssum-Bockhorn	2024	Profile B

7. K-means

unemployment cluster statistics, 2024; percent

cluster	n	unemployment mean	unemployment std	unemployment min	unemployment max
Profile A	51	8.449	2.904	2.590	14.162
Profile B	28	18.077	3.379	13.601	27.541

7. K-means

employment cluster statistics, 2024; percent

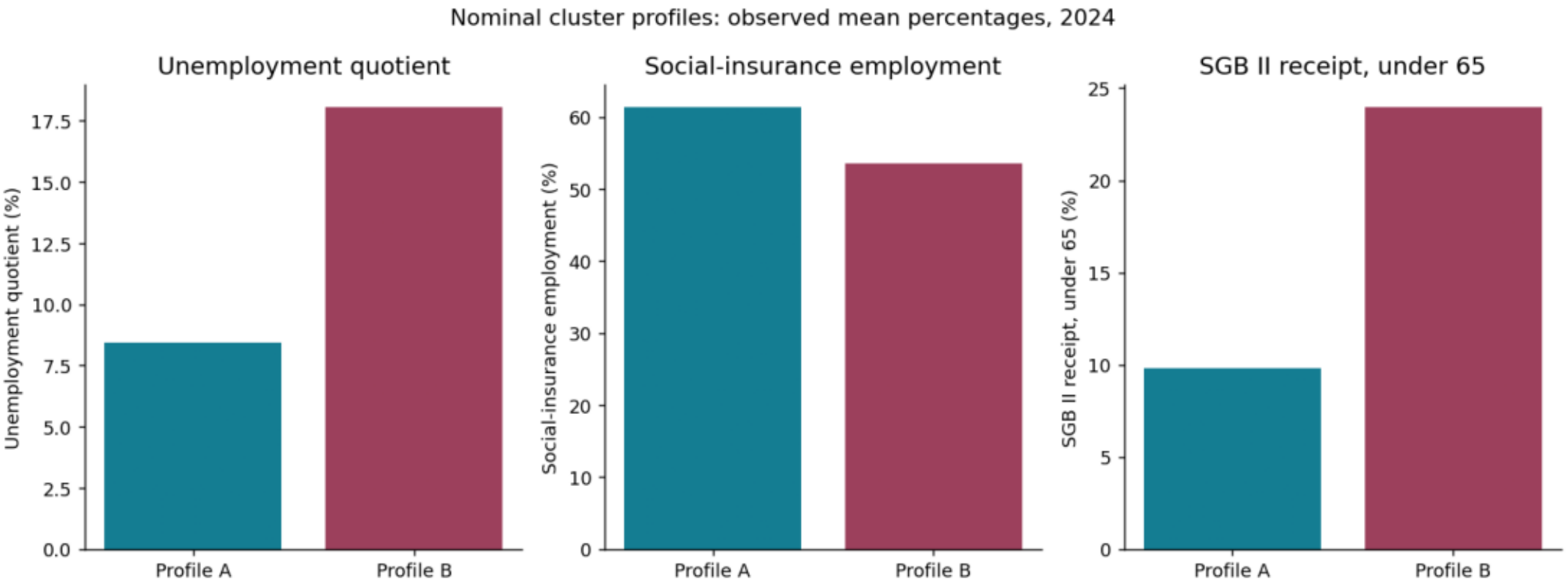
cluster	n	employment mean	employment std	employment min	employment max
Profile A	51	61.404	3.583	51.704	72.714
Profile B	28	53.579	4.096	44.159	59.078

7. K-means

benefit_receipt cluster statistics, 2024; percent

cluster	n	benefit receipt mean	benefit receipt std	benefit receipt min	benefit receipt max
Profile A	51	9.823	5.034	1.702	20.103
Profile B	28	23.989	4.712	17.436	34.622

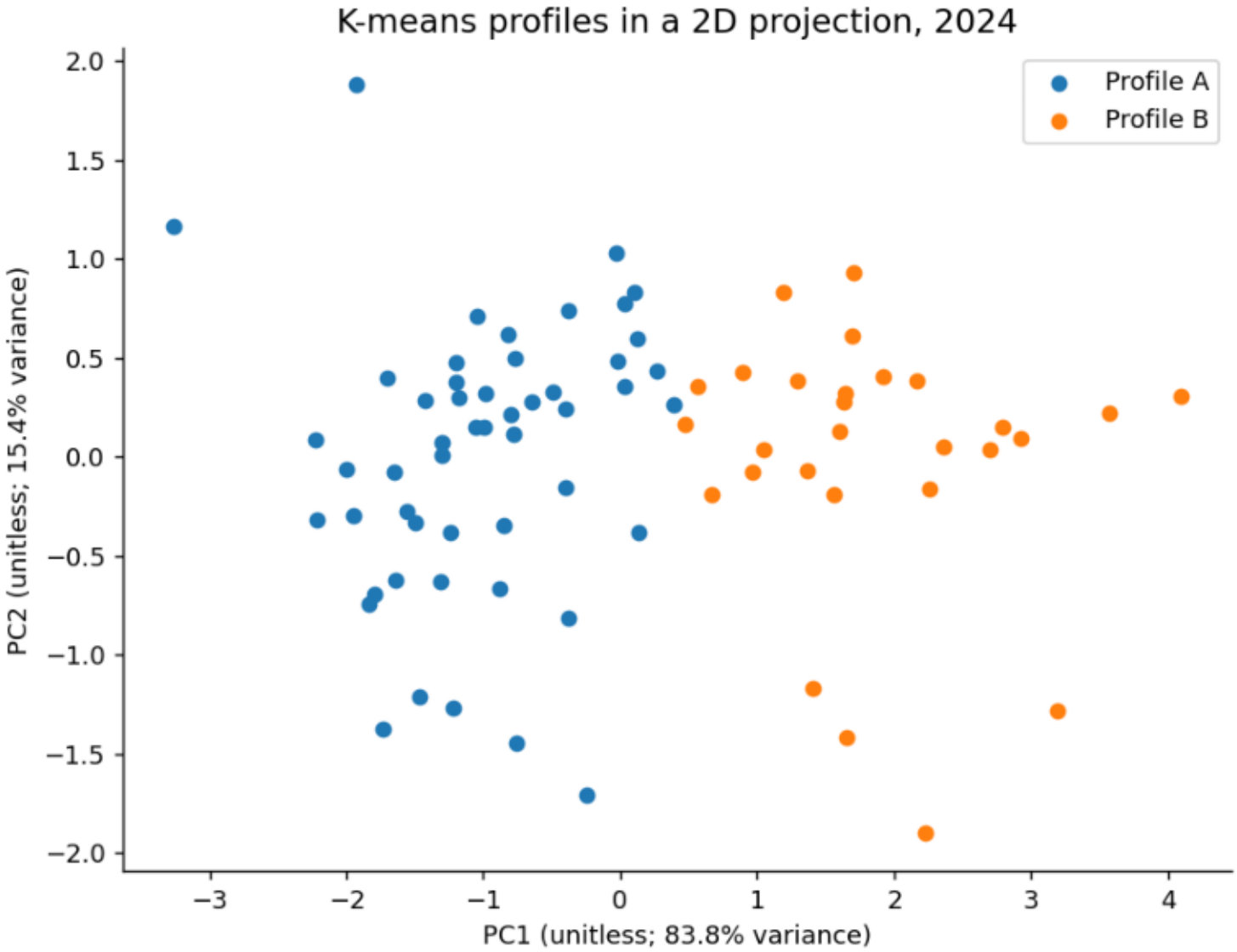
7. K-means



Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Labels are arbitrary; clusters are profiles, not ranks. Means are unweighted.

2024 cluster mean percentages; labels are nominal.

7. K-means



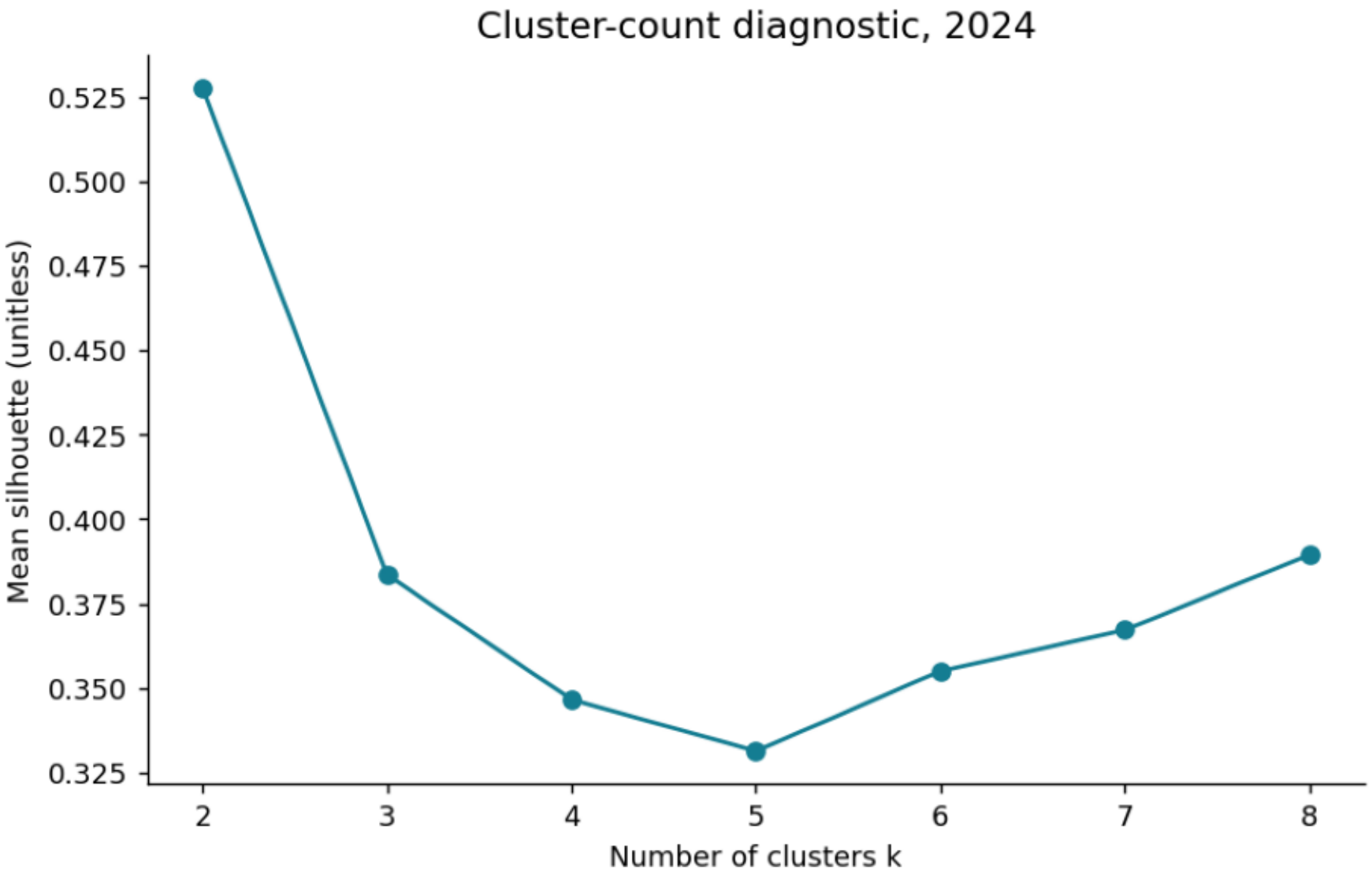
2024 PCA projection; standardized input; unitless coordinates.

7. K-means

PCA explained variance ratios

component	explained variance ratio
PC1	0.838
PC2	0.154

7. K-means



Source: Statistisches Landesamt Bremen, supplied Datentabelle.xls
Highest silhouette selected among solutions with minimum cluster size at least three.

Silhouette by tested cluster count; unitless.

8. Ranking Comparison

Same 2024 row set and three indicators; rank 1 = greatest measured disadvantage.

8. Ranking Comparison

Rank agreement statistics, 2024

spearman	kendall tau b	n
0.974	0.883	79

8. Ranking Comparison

All pairwise rank differences, 2024

geo code	geo name	year	composite rank	topsis rank	rank difference topsis minus composite	absolute rank difference
252	Rablinghausen	2024	73.000	50.000	-23.000	23.000
515	St. Magnus	2024	38.000	57.000	19.000	19.000
342	Lehe	2024	49.000	63.000	14.000	14.000
525	Fähr-Lobbendorf	2024	10.000	20.000	10.000	10.000
361	Oberneuland	2024	60.000	69.000	9.000	9.000
385	Mahndorf	2024	54.000	45.000	-9.000	9.000
437	Ueberseestadt	2024	16.000	24.000	8.000	8.000
311	Steintor	2024	51.000	58.000	7.000	7.000
113	Ostertor	2024	42.000	49.000	7.000	7.000
334	Neue Vahr Südwest	2024	36.000	29.000	-7.000	7.000
111	Altstadt	2024	47.000	41.000	-6.000	6.000
218	Huckelriede	2024	19.000	25.000	6.000	6.000
322	Bürgerpark	2024	65.000	71.000	6.000	6.000
335	Neue Vahr Südost	2024	22.000	16.000	-6.000	6.000

8. Ranking Comparison

All pairwise rank differences, 2024

geo code	geo name	year	composite rank	topsis rank	rank difference topsis minus composite	absolute rank difference
213	Neustadt	2024	50.000	56.000	6.000	6.000
535	Rekum	2024	70.000	64.000	-6.000	6.000
514	Lesum	2024	33.000	38.000	5.000	5.000
511	Burg-Grambke	2024	32.000	27.000	-5.000	5.000
422	Findorff-Bürger weide	2024	57.000	52.000	-5.000	5.000
384	Arbergen	2024	64.000	59.000	-5.000	5.000
351	Borgfeld	2024	71.000	76.000	5.000	5.000
244	Grolland	2024	76.000	72.000	-4.000	4.000
375	Blockdiek	2024	25.000	21.000	-4.000	4.000
383	Hemelingen	2024	21.000	17.000	-4.000	4.000
232	Arsten	2024	58.000	54.000	-4.000	4.000
217	Neuenland	2024	13.000	9.000	-4.000	4.000
331	Gartenstadt Vahr	2024	52.000	48.000	-4.000	4.000
312	Fesenfeld	2024	59.000	62.000	3.000	3.000

8. Ranking Comparison

All pairwise rank differences, 2024

geo code	geo name	year	composite rank	topsis rank	rank difference topsis minus composite	absolute rank difference
534	Farge	2024	31.000	34.000	3.000	3.000
522	Grohn	2024	3.000	6.000	3.000	3.000
382	Hastedt	2024	43.000	40.000	-3.000	3.000
432	Steffensweg	2024	37.000	35.000	-2.000	2.000
532	Rönnebeck	2024	28.000	30.000	2.000	2.000
433	Westend	2024	26.000	28.000	2.000	2.000
341	Horn	2024	72.000	74.000	2.000	2.000
325	Radio Bremen	2024	63.000	65.000	2.000	2.000
216	Buntentor	2024	55.000	53.000	-2.000	2.000
233	Kattenturm	2024	17.000	15.000	-2.000	2.000
215	Gartenstadt Süd	2024	39.000	37.000	-2.000	2.000
212	Hohentor	2024	29.000	31.000	2.000	2.000
371	Ellener Feld	2024	53.000	51.000	-2.000	2.000
314	Hulsberg	2024	48.000	46.000	-2.000	2.000
243	Kirchhuchting	2024	8.000	10.000	2.000	2.000
251	Woltmershausen	2024	30.000	32.000	2.000	2.000

8. Ranking Comparison

All pairwise rank differences, 2024

geo code	geo name	year	composite rank	topsis rank	rank difference topsis minus composite	absolute rank difference
323	Barkhof	2024	75.000	73.000	-2.000	2.000
112	Bahnhofsvorstadt	2024	9.000	8.000	-1.000	1.000
214	Südervorstadt	2024	44.000	43.000	-1.000	1.000
523	Schönebeck	2024	41.000	42.000	1.000	1.000
234	Kattenesch	2024	56.000	55.000	-1.000	1.000
211	Alte Neustadt	2024	45.000	44.000	-1.000	1.000
531	Blumenthal	2024	4.000	3.000	-1.000	1.000
431	Utbremen	2024	34.000	33.000	-1.000	1.000
441	Lindenhof	2024	5.000	4.000	-1.000	1.000
435	Osterfeuerberg	2024	24.000	23.000	-1.000	1.000
434	Walle	2024	27.000	26.000	-1.000	1.000
324	Riensberg	2024	67.000	68.000	1.000	1.000
326	Schwachhausen	2024	74.000	75.000	1.000	1.000
241	Mittelshuchting	2024	14.000	13.000	-1.000	1.000

8. Ranking Comparison

All pairwise rank differences, 2024

geo code	geo name	year	composite rank	topsis rank	rank difference topsis minus composite	absolute rank difference
321	Neu-Schwachhausen	2024	68.000	67.000	-1.000	1.000
343	Lehesterdeich	2024	61.000	60.000	-1.000	1.000
327	Gete	2024	69.000	70.000	1.000	1.000
372	Ellenerbrok-Schevemoor	2024	35.000	36.000	1.000	1.000
373	Tenever	2024	6.000	5.000	-1.000	1.000
421	Regensburger Straße	2024	46.000	47.000	1.000	1.000
524	Aumund-Hammersbeck	2024	23.000	22.000	-1.000	1.000
521	Vegesack	2024	20.000	19.000	-1.000	1.000
513	Burgdamm	2024	15.000	14.000	-1.000	1.000
423	Weidedamm	2024	62.000	61.000	-1.000	1.000
381	Sebaldsbrück	2024	40.000	39.000	-1.000	1.000
231	Habenhausen	2024	78.000	78.000	0.000	0.000
242	Sodenmatt	2024	11.000	11.000	0.000	0.000

8. Ranking Comparison

All pairwise rank differences, 2024

geo code	geo name	year	composite rank	topsis rank	rank difference topsis minus composite	absolute rank difference
374	Osterholz	2024	77.000	77.000	0.000	0.000
332	Neue Vahr Nord	2024	12.000	12.000	0.000	0.000
261	Seehausen	2024	79.000	79.000	0.000	0.000
313	Peterswerder	2024	66.000	66.000	0.000	0.000
442	Gröpelingen	2024	2.000	2.000	0.000	0.000
443	Ohlenhof	2024	1.000	1.000	0.000	0.000
445	Oslebshausen	2024	18.000	18.000	0.000	0.000
533	Lüssum-Bockhorn	2024	7.000	7.000	0.000	0.000

9. Sensitivity and Stability

Fixed ranking sample; alternative weights and feature sets; 30 initialization and 30 bootstrap refits.

9. Sensitivity and Stability

Named sensitivity scenarios: feature sets and weight vectors

scenario	method	features	weights
equal	composite	unemployment; employment; benefit_receipt	0.33333333; 0.33333333; 0.33333333
equal	topsis	unemployment; employment; benefit_receipt	0.33333333; 0.33333333; 0.33333333
domain_balanced	composite	unemployment; employment; benefit_receipt	0.25000000; 0.25000000; 0.50000000
domain_balanced	topsis	unemployment; employment; benefit_receipt	0.25000000; 0.25000000; 0.50000000
unemployment_emphasis	composite	unemployment; employment; benefit_receipt	0.50000000; 0.25000000; 0.25000000
unemployment_emphasis	topsis	unemployment; employment; benefit_receipt	0.50000000; 0.25000000; 0.25000000
employment_emphasis	composite	unemployment; employment; benefit_receipt	0.25000000; 0.50000000; 0.25000000
employment_emphasis	topsis	unemployment; employment; benefit_receipt	0.25000000; 0.50000000; 0.25000000
omit_unemployment	composite	employment; benefit_receipt	0.50000000; 0.50000000
omit_unemployment	topsis	employment; benefit_receipt	0.50000000; 0.50000000
omit_employment	composite	unemployment; benefit_receipt	0.50000000; 0.50000000
omit_employment	topsis	unemployment; benefit_receipt	0.50000000; 0.50000000
omit_benefit_receipt	composite	unemployment; employment	0.50000000; 0.50000000
omit_benefit_receipt	topsis	unemployment; employment	0.50000000; 0.50000000
include_child_benefits	composite	unemployment; employment; benefit_receipt; child_benefits	0.25000000; 0.25000000; 0.25000000; 0.25000000
include_child_benefits	topsis	unemployment; employment; benefit_receipt; child_benefits	0.25000000; 0.25000000; 0.25000000; 0.25000000

9. Sensitivity and Stability

Named scenario rank robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
equal	composite	1.000	0.000	0.000	10	79
equal	topsis	1.000	0.000	0.000	10	79
domain_balanced	composite	0.994	1.722	8.000	9	79
domain_balanced	topsis	0.995	1.468	10.000	9	79
unemployment_emphasis	composite	0.997	1.190	7.000	10	79
unemployment_emphasis	topsis	0.997	1.241	5.000	10	79
employment_emphasis	composite	0.986	2.759	11.000	9	79
employment_emphasis	topsis	0.996	1.266	8.000	9	79
omit_unemployment	composite	0.991	2.278	9.000	9	79
omit_unemployment	topsis	0.993	1.873	10.000	9	79
omit_employment	composite	0.968	3.873	24.000	8	79
omit_employment	topsis	0.999	0.684	4.000	10	79
omit_benefit_receipt	composite	0.972	4.000	13.000	10	79
omit_benefit_receipt	topsis	0.992	2.127	12.000	10	79
include_child_benefits	composite	0.993	1.975	7.000	9	79
include_child_benefits	topsis	0.994	1.772	7.000	9	79

9. Sensitivity and Stability

200 constrained random weight scenarios per method: summary

method	scenarios	minimum rank correlation	mean rank correlation	maximum rank movement	minimum top10 overlap
composite	200	0.962	0.991	18.000	9
topsis	200	0.981	0.996	18.000	8

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_000	composite	0.991	2.152	8.000	10	79
plausible_random_000	topsis	0.997	1.114	8.000	10	79
plausible_random_001	composite	0.974	3.848	15.000	9	79
plausible_random_001	topsis	0.989	2.203	14.000	10	79
plausible_random_002	composite	0.998	0.937	5.000	9	79
plausible_random_002	topsis	0.999	0.785	5.000	9	79
plausible_random_003	composite	0.999	0.481	4.000	10	79
plausible_random_003	topsis	0.998	0.886	5.000	9	79
plausible_random_004	composite	0.990	2.203	11.000	9	79
plausible_random_004	topsis	0.994	1.722	11.000	8	79
plausible_random_005	composite	0.987	2.329	15.000	9	79
plausible_random_005	topsis	0.998	1.089	4.000	10	79
plausible_random_006	composite	0.970	4.127	15.000	9	79
plausible_random_006	topsis	0.986	2.380	16.000	10	79
plausible_random_007	composite	0.997	1.266	7.000	9	79
plausible_random_007	topsis	0.997	1.241	5.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_008	composite	0.992	1.975	10.000	9	79
plausible_random_008	topsis	0.997	1.316	8.000	9	79
plausible_random_009	composite	0.970	4.101	16.000	9	79
plausible_random_009	topsis	0.991	2.000	10.000	9	79
plausible_random_010	composite	0.994	1.620	10.000	9	79
plausible_random_010	topsis	0.997	1.342	5.000	10	79
plausible_random_011	composite	0.998	1.013	4.000	10	79
plausible_random_011	topsis	0.998	0.987	7.000	10	79
plausible_random_012	composite	1.000	0.278	2.000	10	79
plausible_random_012	topsis	0.998	0.911	5.000	10	79
plausible_random_013	composite	0.996	1.291	7.000	10	79
plausible_random_013	topsis	0.994	1.696	10.000	9	79
plausible_random_014	composite	0.993	1.797	11.000	9	79
plausible_random_014	topsis	0.997	1.418	5.000	10	79
plausible_random_015	composite	0.999	0.481	3.000	10	79
plausible_random_015	topsis	0.998	0.861	5.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_016	composite	0.994	1.671	10.000	9	79
plausible_random_016	topsis	0.999	0.608	3.000	10	79
plausible_random_017	composite	0.998	1.114	4.000	10	79
plausible_random_017	topsis	0.996	1.392	8.000	10	79
plausible_random_018	composite	0.995	1.544	9.000	9	79
plausible_random_018	topsis	0.997	1.342	5.000	10	79
plausible_random_019	composite	0.989	2.228	15.000	9	79
plausible_random_019	topsis	0.997	1.367	5.000	10	79
plausible_random_020	composite	0.977	3.620	13.000	9	79
plausible_random_020	topsis	0.991	2.051	14.000	10	79
plausible_random_021	composite	0.998	0.962	5.000	10	79
plausible_random_021	topsis	0.997	1.342	6.000	10	79
plausible_random_022	composite	0.976	3.671	15.000	9	79
plausible_random_022	topsis	0.993	1.747	10.000	9	79
plausible_random_023	composite	0.997	1.063	5.000	9	79
plausible_random_023	topsis	0.999	0.709	5.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_024	composite	0.999	0.709	5.000	10	79
plausible_random_024	topsis	0.996	1.418	10.000	9	79
plausible_random_025	composite	0.977	3.620	13.000	9	79
plausible_random_025	topsis	0.990	2.127	14.000	10	79
plausible_random_026	composite	0.992	1.924	10.000	9	79
plausible_random_026	topsis	0.997	1.316	8.000	9	79
plausible_random_027	composite	0.987	2.456	13.000	9	79
plausible_random_027	topsis	0.995	1.646	10.000	8	79
plausible_random_028	composite	0.982	3.063	13.000	9	79
plausible_random_028	topsis	0.995	1.494	7.000	9	79
plausible_random_029	composite	0.998	0.987	4.000	10	79
plausible_random_029	topsis	0.994	1.747	10.000	10	79
plausible_random_030	composite	0.988	2.430	13.000	9	79
plausible_random_030	topsis	0.995	1.646	11.000	8	79
plausible_random_031	composite	0.998	0.886	3.000	9	79
plausible_random_031	topsis	0.997	1.241	9.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_032	composite	0.981	3.241	12.000	9	79
plausible_random_032	topsis	0.991	2.127	12.000	10	79
plausible_random_033	composite	0.995	1.494	9.000	9	79
plausible_random_033	topsis	0.997	1.418	5.000	10	79
plausible_random_034	composite	0.990	2.228	11.000	9	79
plausible_random_034	topsis	0.995	1.620	10.000	8	79
plausible_random_035	composite	0.999	0.734	3.000	9	79
plausible_random_035	topsis	0.999	0.709	5.000	9	79
plausible_random_036	composite	0.994	1.747	7.000	10	79
plausible_random_036	topsis	0.998	0.709	6.000	9	79
plausible_random_037	composite	0.999	0.557	3.000	10	79
plausible_random_037	topsis	1.000	0.177	1.000	10	79
plausible_random_038	composite	0.997	1.241	5.000	10	79
plausible_random_038	topsis	0.997	1.316	8.000	10	79
plausible_random_039	composite	1.000	0.127	1.000	10	79
plausible_random_039	topsis	0.999	0.481	3.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_040	composite	0.996	1.367	5.000	10	79
plausible_random_040	topsis	0.997	1.266	8.000	10	79
plausible_random_041	composite	0.997	1.165	5.000	9	79
plausible_random_041	topsis	0.998	0.962	5.000	9	79
plausible_random_042	composite	0.998	0.987	4.000	10	79
plausible_random_042	topsis	0.994	1.722	10.000	10	79
plausible_random_043	composite	0.997	1.165	5.000	9	79
plausible_random_043	topsis	0.994	1.747	11.000	8	79
plausible_random_044	composite	0.994	1.772	9.000	9	79
plausible_random_044	topsis	0.995	1.570	10.000	9	79
plausible_random_045	composite	0.989	2.203	13.000	9	79
plausible_random_045	topsis	0.999	0.456	3.000	10	79
plausible_random_046	composite	0.994	1.823	6.000	10	79
plausible_random_046	topsis	0.995	1.722	10.000	10	79
plausible_random_047	composite	0.993	1.899	9.000	9	79
plausible_random_047	topsis	0.995	1.671	10.000	8	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_048	composite	0.985	2.557	15.000	9	79
plausible_random_048	topsis	0.998	1.139	5.000	9	79
plausible_random_049	composite	0.992	2.127	8.000	10	79
plausible_random_049	topsis	0.997	1.089	7.000	10	79
plausible_random_050	composite	0.991	2.076	11.000	9	79
plausible_random_050	topsis	0.997	1.190	7.000	9	79
plausible_random_051	composite	0.999	0.380	3.000	10	79
plausible_random_051	topsis	0.999	0.759	5.000	9	79
plausible_random_052	composite	0.997	1.190	4.000	10	79
plausible_random_052	topsis	0.996	1.392	8.000	10	79
plausible_random_053	composite	0.998	0.987	6.000	9	79
plausible_random_053	topsis	0.999	0.506	3.000	10	79
plausible_random_054	composite	0.998	1.063	5.000	9	79
plausible_random_054	topsis	0.994	1.671	10.000	8	79
plausible_random_055	composite	0.981	3.215	13.000	9	79
plausible_random_055	topsis	0.994	1.519	9.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_056	composite	0.997	1.089	6.000	9	79
plausible_random_056	topsis	0.999	0.430	2.000	10	79
plausible_random_057	composite	0.984	2.582	15.000	9	79
plausible_random_057	topsis	0.998	1.139	5.000	9	79
plausible_random_058	composite	0.978	3.443	14.000	9	79
plausible_random_058	topsis	0.993	1.646	10.000	9	79
plausible_random_059	composite	0.997	1.165	6.000	10	79
plausible_random_059	topsis	0.996	1.392	9.000	9	79
plausible_random_060	composite	0.979	3.392	13.000	9	79
plausible_random_060	topsis	0.991	2.101	13.000	10	79
plausible_random_061	composite	0.995	1.443	8.000	9	79
plausible_random_061	topsis	0.999	0.785	4.000	10	79
plausible_random_062	composite	0.995	1.519	8.000	10	79
plausible_random_062	topsis	0.994	1.722	10.000	9	79
plausible_random_063	composite	0.973	3.873	15.000	9	79
plausible_random_063	topsis	0.992	1.924	10.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_064	composite	0.993	1.975	8.000	10	79
plausible_random_064	topsis	0.998	0.937	7.000	10	79
plausible_random_065	composite	0.994	1.570	9.000	9	79
plausible_random_065	topsis	0.998	1.013	4.000	10	79
plausible_random_066	composite	0.996	1.443	7.000	10	79
plausible_random_066	topsis	0.996	1.392	8.000	9	79
plausible_random_067	composite	0.992	2.127	8.000	10	79
plausible_random_067	topsis	0.997	0.987	7.000	10	79
plausible_random_068	composite	0.997	1.089	5.000	10	79
plausible_random_068	topsis	0.998	0.861	7.000	10	79
plausible_random_069	composite	0.997	1.241	5.000	9	79
plausible_random_069	topsis	0.997	1.165	7.000	9	79
plausible_random_070	composite	0.971	4.051	15.000	9	79
plausible_random_070	topsis	0.987	2.380	16.000	10	79
plausible_random_071	composite	0.987	2.354	15.000	9	79
plausible_random_071	topsis	0.999	0.658	3.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_072	composite	0.994	1.722	8.000	9	79
plausible_random_072	topsis	0.993	1.873	11.000	8	79
plausible_random_073	composite	0.991	2.203	9.000	10	79
plausible_random_073	topsis	0.997	0.962	7.000	9	79
plausible_random_074	composite	0.992	2.025	9.000	9	79
plausible_random_074	topsis	0.994	1.671	11.000	8	79
plausible_random_075	composite	0.998	0.987	4.000	9	79
plausible_random_075	topsis	0.995	1.468	10.000	9	79
plausible_random_076	composite	0.982	2.734	16.000	9	79
plausible_random_076	topsis	0.998	0.987	5.000	9	79
plausible_random_077	composite	0.996	1.342	6.000	10	79
plausible_random_077	topsis	0.999	0.684	4.000	9	79
plausible_random_078	composite	0.995	1.671	7.000	10	79
plausible_random_078	topsis	0.997	1.215	7.000	9	79
plausible_random_079	composite	0.998	0.911	5.000	10	79
plausible_random_079	topsis	0.998	0.835	4.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_080	composite	0.999	0.354	4.000	10	79
plausible_random_080	topsis	1.000	0.101	1.000	10	79
plausible_random_081	composite	0.999	0.354	4.000	10	79
plausible_random_081	topsis	1.000	0.101	1.000	10	79
plausible_random_082	composite	0.980	2.937	18.000	9	79
plausible_random_082	topsis	0.997	1.266	7.000	9	79
plausible_random_083	composite	1.000	0.127	1.000	10	79
plausible_random_083	topsis	0.999	0.456	3.000	10	79
plausible_random_084	composite	0.970	4.076	16.000	9	79
plausible_random_084	topsis	0.987	2.430	15.000	10	79
plausible_random_085	composite	0.999	0.380	2.000	10	79
plausible_random_085	topsis	0.999	0.608	3.000	10	79
plausible_random_086	composite	0.998	0.734	6.000	10	79
plausible_random_086	topsis	0.995	1.570	10.000	9	79
plausible_random_087	composite	0.999	0.582	5.000	10	79
plausible_random_087	topsis	0.998	0.962	6.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_088	composite	0.983	2.759	15.000	9	79
plausible_random_088	topsis	0.995	1.646	10.000	9	79
plausible_random_089	composite	0.998	0.886	3.000	9	79
plausible_random_089	topsis	0.997	1.241	8.000	9	79
plausible_random_090	composite	0.989	2.481	9.000	10	79
plausible_random_090	topsis	0.995	1.418	10.000	10	79
plausible_random_091	composite	0.991	2.025	10.000	9	79
plausible_random_091	topsis	0.996	1.468	9.000	9	79
plausible_random_092	composite	0.993	1.873	8.000	10	79
plausible_random_092	topsis	0.998	1.013	7.000	9	79
plausible_random_093	composite	0.990	2.101	12.000	9	79
plausible_random_093	topsis	0.999	0.405	3.000	10	79
plausible_random_094	composite	0.993	1.924	8.000	10	79
plausible_random_094	topsis	0.996	1.342	8.000	10	79
plausible_random_095	composite	0.968	4.228	17.000	9	79
plausible_random_095	topsis	0.989	2.152	13.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_096	composite	0.989	2.481	10.000	10	79
plausible_random_096	topsis	0.996	1.190	8.000	10	79
plausible_random_097	composite	0.996	1.392	6.000	10	79
plausible_random_097	topsis	0.998	0.835	5.000	9	79
plausible_random_098	composite	0.997	1.291	5.000	9	79
plausible_random_098	topsis	0.996	1.342	9.000	9	79
plausible_random_099	composite	0.990	2.354	9.000	10	79
plausible_random_099	topsis	0.997	1.190	8.000	10	79
plausible_random_100	composite	0.993	1.823	9.000	9	79
plausible_random_100	topsis	0.995	1.595	10.000	8	79
plausible_random_101	composite	0.999	0.405	3.000	10	79
plausible_random_101	topsis	1.000	0.354	3.000	9	79
plausible_random_102	composite	0.995	1.544	8.000	9	79
plausible_random_102	topsis	0.998	1.038	5.000	9	79
plausible_random_103	composite	0.987	2.709	10.000	10	79
plausible_random_103	topsis	0.993	1.949	12.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_104	composite	0.991	2.278	9.000	10	79
plausible_random_104	topsis	0.996	1.392	9.000	10	79
plausible_random_105	composite	0.996	1.570	5.000	10	79
plausible_random_105	topsis	0.994	1.696	10.000	10	79
plausible_random_106	composite	0.997	1.139	7.000	9	79
plausible_random_106	topsis	1.000	0.304	2.000	10	79
plausible_random_107	composite	0.997	1.190	7.000	9	79
plausible_random_107	topsis	0.999	0.684	4.000	10	79
plausible_random_108	composite	0.998	0.962	4.000	10	79
plausible_random_108	topsis	0.994	1.696	10.000	10	79
plausible_random_109	composite	0.988	2.354	13.000	9	79
plausible_random_109	topsis	0.997	1.190	7.000	9	79
plausible_random_110	composite	0.986	2.734	11.000	9	79
plausible_random_110	topsis	0.996	1.443	8.000	9	79
plausible_random_111	composite	0.993	1.899	8.000	10	79
plausible_random_111	topsis	0.997	1.139	7.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_112	composite	0.986	2.734	11.000	9	79
plausible_random_112	topsis	0.995	1.620	8.000	9	79
plausible_random_113	composite	0.998	0.962	5.000	10	79
plausible_random_113	topsis	0.999	0.658	6.000	10	79
plausible_random_114	composite	0.995	1.646	8.000	9	79
plausible_random_114	topsis	0.995	1.570	10.000	9	79
plausible_random_115	composite	0.974	3.848	15.000	9	79
plausible_random_115	topsis	0.989	2.228	14.000	10	79
plausible_random_116	composite	0.999	0.658	3.000	10	79
plausible_random_116	topsis	0.997	1.266	7.000	10	79
plausible_random_117	composite	0.998	0.987	4.000	9	79
plausible_random_117	topsis	0.995	1.468	10.000	9	79
plausible_random_118	composite	0.999	0.684	4.000	10	79
plausible_random_118	topsis	0.996	1.544	8.000	10	79
plausible_random_119	composite	0.979	3.342	14.000	9	79
plausible_random_119	topsis	0.994	1.696	8.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_120	composite	0.998	0.937	5.000	9	79
plausible_random_120	topsis	0.994	1.747	10.000	8	79
plausible_random_121	composite	0.984	2.962	12.000	9	79
plausible_random_121	topsis	0.995	1.367	9.000	9	79
plausible_random_122	composite	0.979	3.392	13.000	9	79
plausible_random_122	topsis	0.991	2.152	13.000	10	79
plausible_random_123	composite	0.999	0.582	2.000	10	79
plausible_random_123	topsis	0.999	0.633	3.000	10	79
plausible_random_124	composite	0.995	1.570	7.000	10	79
plausible_random_124	topsis	0.996	1.443	8.000	9	79
plausible_random_125	composite	0.987	2.329	15.000	9	79
plausible_random_125	topsis	0.999	0.532	3.000	10	79
plausible_random_126	composite	0.991	2.000	10.000	9	79
plausible_random_126	topsis	0.997	1.266	7.000	9	79
plausible_random_127	composite	0.999	0.835	3.000	10	79
plausible_random_127	topsis	0.996	1.570	7.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_128	composite	0.989	2.456	9.000	10	79
plausible_random_128	topsis	0.995	1.443	10.000	10	79
plausible_random_129	composite	0.978	3.519	14.000	9	79
plausible_random_129	topsis	0.993	1.696	9.000	9	79
plausible_random_130	composite	0.996	1.266	7.000	10	79
plausible_random_130	topsis	0.995	1.544	10.000	9	79
plausible_random_131	composite	0.998	0.911	5.000	10	79
plausible_random_131	topsis	0.999	0.608	4.000	9	79
plausible_random_132	composite	0.983	3.063	12.000	9	79
plausible_random_132	topsis	0.995	1.367	9.000	9	79
plausible_random_133	composite	0.997	1.139	5.000	10	79
plausible_random_133	topsis	0.999	0.557	4.000	9	79
plausible_random_134	composite	0.975	3.772	13.000	9	79
plausible_random_134	topsis	0.989	2.228	14.000	10	79
plausible_random_135	composite	0.962	4.633	17.000	9	79
plausible_random_135	topsis	0.981	2.937	18.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_136	composite	0.999	0.582	3.000	10	79
plausible_random_136	topsis	0.997	1.342	7.000	10	79
plausible_random_137	composite	0.990	2.430	9.000	10	79
plausible_random_137	topsis	0.995	1.696	10.000	10	79
plausible_random_138	composite	0.974	3.797	15.000	9	79
plausible_random_138	topsis	0.989	2.228	14.000	10	79
plausible_random_139	composite	0.987	2.608	10.000	10	79
plausible_random_139	topsis	0.995	1.519	10.000	10	79
plausible_random_140	composite	0.992	2.152	7.000	10	79
plausible_random_140	topsis	0.995	1.772	10.000	10	79
plausible_random_141	composite	0.999	0.557	3.000	10	79
plausible_random_141	topsis	0.997	1.291	7.000	10	79
plausible_random_142	composite	0.997	1.139	5.000	9	79
plausible_random_142	topsis	0.998	1.038	6.000	9	79
plausible_random_143	composite	0.982	3.089	12.000	9	79
plausible_random_143	topsis	0.994	1.570	11.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_144	composite	0.987	2.304	15.000	9	79
plausible_random_144	topsis	0.999	0.759	4.000	10	79
plausible_random_145	composite	0.973	3.899	16.000	9	79
plausible_random_145	topsis	0.991	2.000	12.000	9	79
plausible_random_146	composite	0.989	2.430	9.000	10	79
plausible_random_146	topsis	0.993	1.949	10.000	10	79
plausible_random_147	composite	0.998	0.987	3.000	10	79
plausible_random_147	topsis	0.994	1.722	10.000	10	79
plausible_random_148	composite	0.998	0.759	5.000	10	79
plausible_random_148	topsis	0.998	0.937	5.000	9	79
plausible_random_149	composite	1.000	0.076	1.000	10	79
plausible_random_149	topsis	1.000	0.025	1.000	10	79
plausible_random_150	composite	0.999	0.684	5.000	10	79
plausible_random_150	topsis	0.995	1.443	10.000	9	79
plausible_random_151	composite	0.970	4.051	16.000	9	79
plausible_random_151	topsis	0.990	2.025	12.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_152	composite	0.996	1.241	8.000	10	79
plausible_random_152	topsis	0.994	1.747	10.000	9	79
plausible_random_153	composite	0.997	1.215	5.000	10	79
plausible_random_153	topsis	0.997	1.139	7.000	10	79
plausible_random_154	composite	0.995	1.696	7.000	9	79
plausible_random_154	topsis	0.994	1.797	11.000	8	79
plausible_random_155	composite	0.987	2.709	11.000	9	79
plausible_random_155	topsis	0.996	1.367	7.000	9	79
plausible_random_156	composite	0.999	0.835	4.000	10	79
plausible_random_156	topsis	0.999	0.481	3.000	10	79
plausible_random_157	composite	0.999	0.709	5.000	9	79
plausible_random_157	topsis	0.995	1.494	10.000	9	79
plausible_random_158	composite	0.996	1.418	6.000	10	79
plausible_random_158	topsis	0.997	1.291	7.000	9	79
plausible_random_159	composite	0.995	1.494	8.000	9	79
plausible_random_159	topsis	0.999	0.380	3.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_160	composite	0.977	3.519	15.000	9	79
plausible_random_160	topsis	0.993	1.823	9.000	9	79
plausible_random_161	composite	0.996	1.266	7.000	9	79
plausible_random_161	topsis	0.999	0.506	3.000	10	79
plausible_random_162	composite	0.989	2.481	9.000	10	79
plausible_random_162	topsis	0.994	1.848	10.000	10	79
plausible_random_163	composite	0.991	1.975	12.000	9	79
plausible_random_163	topsis	0.999	0.658	3.000	10	79
plausible_random_164	composite	0.991	2.076	11.000	9	79
plausible_random_164	topsis	0.998	0.962	5.000	9	79
plausible_random_165	composite	0.991	2.101	11.000	9	79
plausible_random_165	topsis	0.999	0.582	2.000	9	79
plausible_random_166	composite	0.977	3.595	13.000	9	79
plausible_random_166	topsis	0.990	2.177	13.000	10	79
plausible_random_167	composite	0.996	1.418	6.000	10	79
plausible_random_167	topsis	0.997	1.114	7.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_168	composite	0.999	0.430	2.000	10	79
plausible_random_168	topsis	0.997	1.190	6.000	10	79
plausible_random_169	composite	0.991	1.975	12.000	9	79
plausible_random_169	topsis	0.999	0.759	4.000	10	79
plausible_random_170	composite	0.997	1.038	6.000	9	79
plausible_random_170	topsis	0.999	0.734	4.000	10	79
plausible_random_171	composite	0.999	0.506	4.000	10	79
plausible_random_171	topsis	0.999	0.582	3.000	10	79
plausible_random_172	composite	0.984	2.962	12.000	9	79
plausible_random_172	topsis	0.994	1.747	11.000	10	79
plausible_random_173	composite	0.968	4.177	16.000	9	79
plausible_random_173	topsis	0.988	2.278	13.000	9	79
plausible_random_174	composite	0.997	1.241	4.000	10	79
plausible_random_174	topsis	0.994	1.747	10.000	10	79
plausible_random_175	composite	0.998	0.962	6.000	10	79
plausible_random_175	topsis	0.995	1.494	10.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_176	composite	0.989	2.203	15.000	9	79
plausible_random_176	topsis	0.997	1.418	5.000	10	79
plausible_random_177	composite	0.998	0.962	4.000	10	79
plausible_random_177	topsis	0.996	1.367	8.000	10	79
plausible_random_178	composite	0.991	2.127	9.000	9	79
plausible_random_178	topsis	0.995	1.595	9.000	9	79
plausible_random_179	composite	0.999	0.557	3.000	10	79
plausible_random_179	topsis	0.997	1.266	7.000	10	79
plausible_random_180	composite	0.999	0.709	3.000	10	79
plausible_random_180	topsis	0.997	1.241	6.000	10	79
plausible_random_181	composite	0.993	1.722	10.000	9	79
plausible_random_181	topsis	1.000	0.380	2.000	10	79
plausible_random_182	composite	0.995	1.418	8.000	9	79
plausible_random_182	topsis	0.999	0.709	4.000	10	79
plausible_random_183	composite	0.982	3.114	12.000	9	79
plausible_random_183	topsis	0.992	2.076	12.000	10	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_184	composite	0.986	2.684	11.000	9	79
plausible_random_184	topsis	0.996	1.494	8.000	9	79
plausible_random_185	composite	0.989	2.354	10.000	9	79
plausible_random_185	topsis	0.997	1.266	7.000	9	79
plausible_random_186	composite	0.995	1.544	7.000	10	79
plausible_random_186	topsis	0.997	1.190	7.000	9	79
plausible_random_187	composite	0.999	0.835	3.000	9	79
plausible_random_187	topsis	0.999	0.532	4.000	9	79
plausible_random_188	composite	0.997	1.013	6.000	10	79
plausible_random_188	topsis	0.995	1.494	10.000	9	79
plausible_random_189	composite	0.990	2.329	10.000	10	79
plausible_random_189	topsis	0.997	1.063	8.000	9	79
plausible_random_190	composite	0.999	0.506	4.000	10	79
plausible_random_190	topsis	0.999	0.810	5.000	9	79
plausible_random_191	composite	0.988	2.354	13.000	9	79
plausible_random_191	topsis	0.996	1.468	9.000	9	79

9. Sensitivity and Stability

Every random-weight scenario: calculated robustness statistics

scenario	method	spearman vs baseline	mean absolute rank change	max absolute rank change	top10 overlap	n
plausible_random_192	composite	0.987	2.684	10.000	10	79
plausible_random_192	topsis	0.994	1.620	11.000	10	79
plausible_random_193	composite	0.999	0.810	3.000	9	79
plausible_random_193	topsis	0.998	0.861	5.000	9	79
plausible_random_194	composite	0.999	0.684	4.000	10	79
plausible_random_194	topsis	0.998	0.886	6.000	10	79
plausible_random_195	composite	0.992	1.873	10.000	9	79
plausible_random_195	topsis	0.998	1.089	6.000	9	79
plausible_random_196	composite	0.971	4.025	15.000	9	79
plausible_random_196	topsis	0.987	2.354	15.000	10	79
plausible_random_197	composite	1.000	0.203	1.000	10	79
plausible_random_197	topsis	1.000	0.354	3.000	9	79
plausible_random_198	composite	0.988	2.354	13.000	9	79
plausible_random_198	topsis	0.995	1.595	10.000	9	79
plausible_random_199	composite	0.998	0.962	5.000	9	79
plausible_random_199	topsis	1.000	0.278	2.000	10	79

9. Sensitivity and Stability

Neighborhood rank ranges across random weight scenarios; not confidence intervals

geo code	geo name	min composite	min topsis	max composite	max topsis	median composite	median topsis
111	Altstadt	42.000	40.000	59.000	47.000	48.000	42.000
112	Bahnhofsvorstadt	7.000	7.000	12.000	9.000	10.000	8.000
113	Ostertor	32.000	42.000	48.000	55.000	40.000	48.000
211	Alte Neustadt	43.000	41.000	55.000	47.000	47.000	43.000
212	Hohentor	29.000	27.000	31.000	35.000	29.000	31.000
213	Neustadt	45.000	49.000	53.000	57.000	49.000	55.000
214	Südervorstadt	42.000	41.000	49.000	45.000	44.000	43.000
215	Gartenstadt Süd	38.000	37.000	48.000	41.000	39.500	37.000
216	Buntentor	51.000	51.000	59.000	58.000	55.000	54.000
217	Neuenland	9.000	5.000	21.000	16.000	14.000	10.000
218	Huckelriede	11.000	23.000	25.000	28.000	19.000	25.000
231	Habenhausen	76.000	77.000	78.000	78.000	77.000	78.000
232	Arsten	52.000	52.000	66.000	59.000	58.000	54.000
233	Kattenturm	14.000	14.000	18.000	16.000	16.000	15.000
234	Kattenesch	55.000	53.000	63.000	58.000	57.000	55.000
241	Mittelshuchting	12.000	12.000	19.000	16.000	14.000	13.000

9. Sensitivity and Stability

Neighborhood rank ranges across random weight scenarios; not confidence intervals

geo code	geo name	min composite	min topsis	max composite	max topsis	median composite	median topsis
242	Sodenmatt	10.000	7.000	13.000	11.000	11.000	11.000
243	Kirchhuchting	8.000	8.000	10.000	11.000	9.000	10.000
244	Grolland	72.000	70.000	76.000	76.000	75.000	73.000
251	Woltmershausen	30.000	29.000	33.000	33.000	30.000	32.000
252	Rablinghausen	57.000	49.000	78.000	63.000	74.000	50.000
261	Seehausen	79.000	79.000	79.000	79.000	79.000	79.000
311	Steintor	42.000	52.000	58.000	61.000	51.000	58.000
312	Fesenfeld	54.000	59.000	63.000	64.000	59.000	63.000
313	Peterswerder	61.000	65.000	68.000	70.000	66.000	67.000
314	Hulsberg	46.000	44.000	51.000	49.000	48.000	47.000
321	Neu-Schwachhausen	68.000	65.000	72.000	72.000	69.000	67.000
322	Bürgerpark	49.000	65.000	71.000	75.000	65.000	71.000
323	Barkhof	72.000	71.000	76.000	76.000	74.000	74.000
324	Riensberg	65.000	65.000	69.000	70.000	67.000	68.000
325	Radio Bremen	52.000	62.000	65.000	67.000	62.000	65.000
326	Schwachhausen	70.000	74.000	75.000	76.000	73.000	75.000

9. Sensitivity and Stability

Neighborhood rank ranges across random weight scenarios; not confidence intervals

geo code	geo name	min composite	min topsis	max composite	max topsis	median composite	median topsis
327	Gete	60.000	69.000	71.000	72.000	69.000	70.000
331	Gartenstadt Vahr	49.000	48.000	59.000	51.000	52.000	49.000
332	Neue Vahr Nord	11.000	10.000	19.000	13.000	13.000	12.000
334	Neue Vahr Südwest	29.000	23.000	44.000	33.000	36.000	30.000
335	Neue Vahr Südost	16.000	14.000	25.000	19.000	22.000	16.000
341	Horn	68.000	72.000	74.000	75.000	72.000	73.000
342	Lehe	34.000	50.000	62.000	64.000	46.000	63.000
343	Lehesterdeich	59.000	58.000	64.000	63.000	62.000	59.000
351	Borgfeld	54.000	67.000	77.000	78.000	71.000	75.000
361	Oberneuland	43.000	59.000	69.000	70.000	59.000	69.000
371	Ellener Feld	49.000	47.000	57.000	58.000	53.000	51.000
372	Ellenerbrok-Scheve moor	32.000	34.000	36.000	36.000	35.000	35.000
373	Tenever	4.000	3.000	7.000	8.000	6.000	5.000
374	Osterholz	76.000	73.000	78.000	78.000	77.000	77.000
375	Blockdiek	21.000	20.000	27.000	25.000	25.000	21.000

9. Sensitivity and Stability

Neighborhood rank ranges across random weight scenarios; not confidence intervals

geo code	geo name	min composite	min topsis	max composite	max topsis	median composite	median topsis
381	Sebaldsbrück	39.000	39.000	42.000	40.000	40.000	39.000
382	Hastedt	39.000	37.000	47.000	42.000	43.000	40.000
383	Hemelingen	17.000	17.000	21.000	20.000	21.000	17.000
384	Arbergen	58.000	54.000	72.000	67.000	64.000	59.000
385	Mahndorf	46.000	43.000	67.000	54.000	54.000	46.000
421	Regensburger Straße	45.000	44.000	50.000	49.000	47.000	47.000
422	Findorff-Bürgerweide	51.000	50.000	69.000	56.000	58.000	52.000
423	Weidedamm	60.000	59.000	63.000	64.000	61.000	61.000
431	Utbremen	29.000	22.000	40.000	37.000	34.000	33.000
432	Steffensweg	34.000	34.000	38.000	36.000	37.000	35.500
433	Westend	24.000	26.000	27.000	31.000	26.000	29.000
434	Walle	26.000	24.000	30.000	29.000	27.000	26.000
435	Osterfeuerberg	22.000	21.000	25.000	29.000	24.000	24.000
437	Ueberseestadt	8.000	20.000	22.000	26.000	14.500	23.000

9. Sensitivity and Stability

Neighborhood rank ranges across random weight scenarios; not confidence intervals

geo code	geo name	min composite	min topsis	max composite	max topsis	median composite	median topsis
441	Lindenhof	4.000	3.000	7.000	5.000	5.000	4.000
442	Gröpelingen	2.000	2.000	3.000	2.000	2.000	2.000
443	Ohlenhof	1.000	1.000	1.000	1.000	1.000	1.000
445	Oslebshausen	14.000	16.000	18.000	18.000	17.000	18.000
511	Burg-Grambke	30.000	26.000	39.000	33.000	33.000	30.000
513	Burgdamm	13.000	13.000	18.000	14.000	15.000	13.000
514	Lesum	29.000	36.000	37.000	39.000	32.000	38.000
515	St. Magnus	26.000	39.000	56.000	62.000	38.000	55.000
521	Vegesack	18.000	18.000	21.000	20.000	20.000	19.000
522	Grohn	2.000	5.000	6.000	8.000	3.000	6.000
523	Schönebeck	39.000	40.000	43.000	44.000	40.000	42.000
524	Aumund-Hammersbeck	21.000	21.000	24.000	24.000	23.000	22.000
525	Fähr-Lobbendorf	4.000	12.000	20.000	27.000	9.000	20.000
531	Blumenthal	3.000	3.000	5.000	4.000	4.000	3.000
532	Rönnebeck	27.000	27.000	29.000	32.000	28.000	29.000
533	Lüssum-Bockhorn	7.000	6.000	10.000	9.000	7.000	7.000

9. Sensitivity and Stability

Neighborhood rank ranges across random weight scenarios; not confidence intervals

geo code	geo name	min composite	min topsis	max composite	max topsis	median composite	median topsis
534	Farge	31.000	31.000	36.000	34.000	33.000	32.000
535	Rekum	64.000	61.000	74.000	72.000	70.000	64.000

9. Sensitivity and Stability

Initialization and bootstrap clustering stability: adjusted Rand index

experiment	mean	min	max
bootstrap_neighborhoods	0.892	0.591	1.000
initialization	1.000	1.000	1.000

9. Sensitivity and Stability

All clustering refit stability statistics

experiment	seed	k	adjusted rand index
initialization	0	2	1.000
bootstrap_neighborhoods	0	2	0.949
initialization	1	2	1.000
bootstrap_neighborhoods	1	2	0.591
initialization	2	2	1.000
bootstrap_neighborhoods	2	2	0.949
initialization	3	2	1.000
bootstrap_neighborhoods	3	2	1.000
initialization	4	2	1.000
bootstrap_neighborhoods	4	2	1.000
initialization	5	2	1.000
bootstrap_neighborhoods	5	2	0.949
initialization	6	2	1.000
bootstrap_neighborhoods	6	2	1.000
initialization	7	2	1.000
bootstrap_neighborhoods	7	2	0.949

9. Sensitivity and Stability

All clustering refit stability statistics

experiment	seed	k	adjusted rand index
initialization	8	2	1.000
bootstrap_neighborhoods	8	2	0.899
initialization	9	2	1.000
bootstrap_neighborhoods	9	2	0.949
initialization	10	2	1.000
bootstrap_neighborhoods	10	2	0.591
initialization	11	2	1.000
bootstrap_neighborhoods	11	2	0.899
initialization	12	2	1.000
bootstrap_neighborhoods	12	2	0.899
initialization	13	2	1.000
bootstrap_neighborhoods	13	2	0.851
initialization	14	2	1.000
bootstrap_neighborhoods	14	2	0.899
initialization	15	2	1.000
bootstrap_neighborhoods	15	2	0.899

9. Sensitivity and Stability

All clustering refit stability statistics

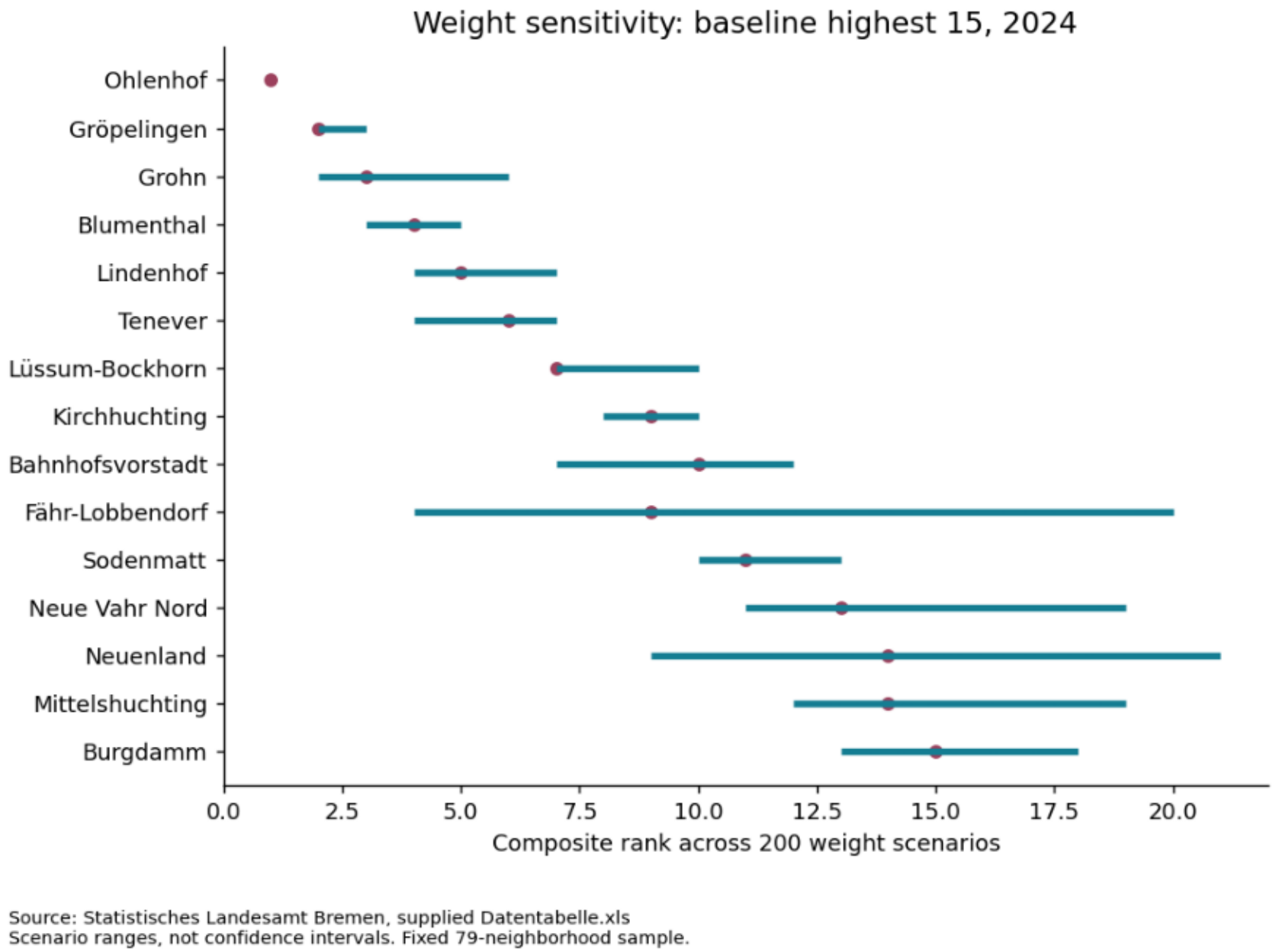
experiment	seed	k	adjusted rand index
initialization	16	2	1.000
bootstrap_neighborhoods	16	2	0.949
initialization	17	2	1.000
bootstrap_neighborhoods	17	2	1.000
initialization	18	2	1.000
bootstrap_neighborhoods	18	2	0.851
initialization	19	2	1.000
bootstrap_neighborhoods	19	2	0.949
initialization	20	2	1.000
bootstrap_neighborhoods	20	2	1.000
initialization	21	2	1.000
bootstrap_neighborhoods	21	2	0.899
initialization	22	2	1.000
bootstrap_neighborhoods	22	2	0.591
initialization	23	2	1.000
bootstrap_neighborhoods	23	2	0.949

9. Sensitivity and Stability

All clustering refit stability statistics

experiment	seed	k	adjusted rand index
initialization	24	2	1.000
bootstrap_neighborhoods	24	2	0.591
initialization	25	2	1.000
bootstrap_neighborhoods	25	2	0.899
initialization	26	2	1.000
bootstrap_neighborhoods	26	2	1.000
initialization	27	2	1.000
bootstrap_neighborhoods	27	2	0.899
initialization	28	2	1.000
bootstrap_neighborhoods	28	2	0.949
initialization	29	2	1.000
bootstrap_neighborhoods	29	2	0.949

9. Sensitivity and Stability

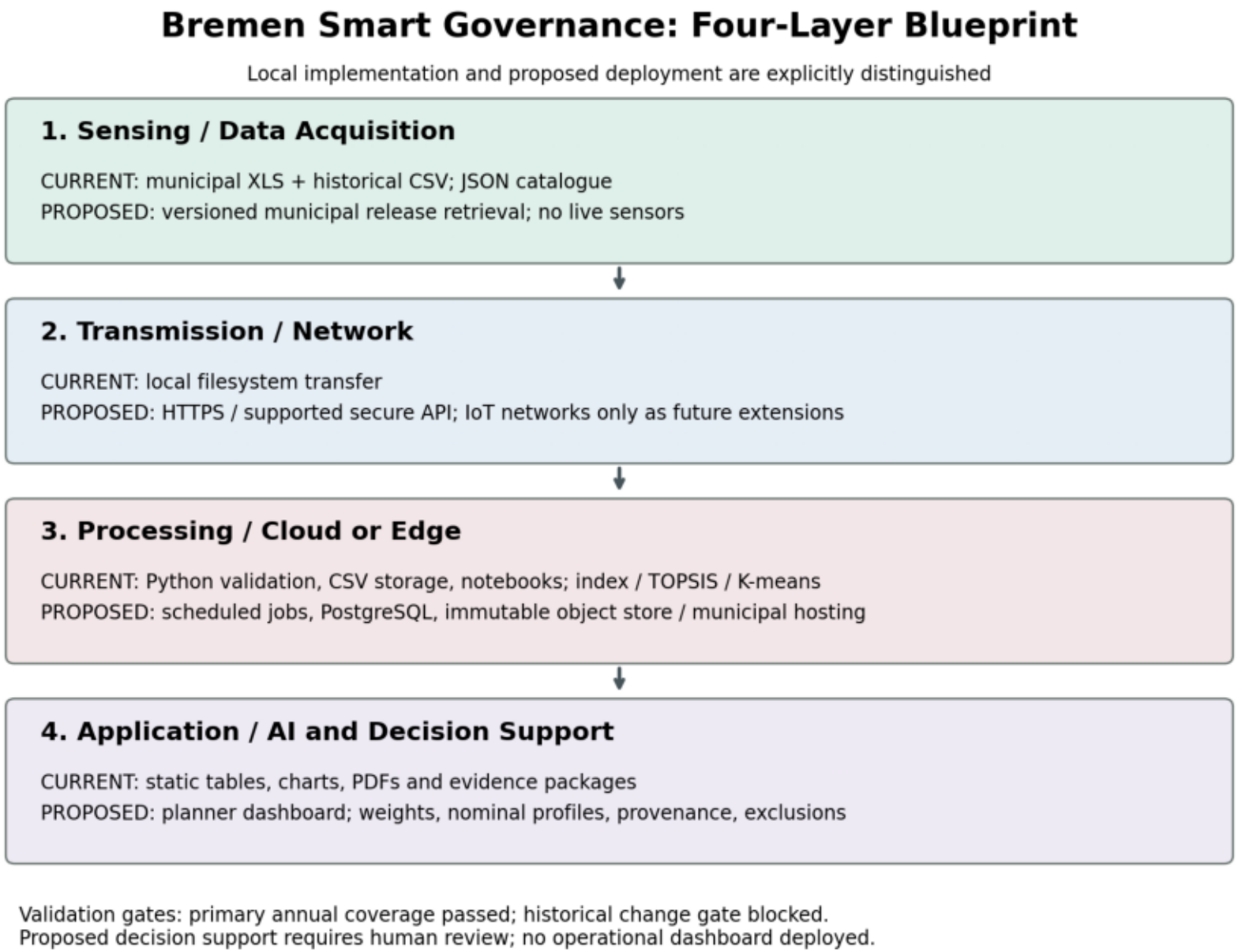


2024 composite rank ranges; 200 weight scenarios; baseline highest 15.

10. Four-Layer Architecture Diagram

Existing architecture figure.

10. Four-Layer Architecture Diagram



Existing four-layer architecture diagram.

11. Sources and Reproducibility

Saved results, source references, calculation settings and script links.

11. Sources and Reproducibility

Calculation settings

Calculation	Setting
Primary inputs	2024; complete cases; unemployment, employment, benefit_receipt
Composite	Min-max; employment reversed; equal weights; descending minimum-tie rank
TOPSIS	Vector normalization; equal weights; Euclidean distances; high unemployment/benefits and low employment ideal
K-means	StandardScaler; k=2..8; highest silhouette with minimum cluster size 3; n_init=30; seed=42
Random weights	200 draws; Dirichlet(3,3,3); weights constrained to [0.10,0.60]; seed=2024
Cluster stability	30 initialization seeds; 30 bootstrap neighborhood refits; ARI on fixed full sample
Display precision	3 decimal places; stored CSV precision unchanged

11. Sources and Reproducibility

Dataset references

Source	Local file	Reference
Primary statistics	Datentabelle.xls	https://www.statistik.bremen.de/datenangebote/bremer-ortsteilatlas-15228
Historical CSV statistics	22 ST*.csv files	data_inventory.csv
Catalogue metadata	bremen_data.json	data/processed/bremen_open_data_catalog.csv
Raw-source checksums	analysis/source_manifest.csv	SHA-256

11. Sources and Reproducibility

Scripts used

Output	Script
Preparation	src/prepare_data.py
Analysis-ready dataset	src/build_analysis_dataset.py
Descriptive statistics	src/descriptive_analysis.py
Composite	src/composite_index.py
TOPSIS	src/topsis.py
K-means	src/clustering.py
Sensitivity	src/sensitivity_analysis.py
Figures and packages	src/package_results.py
Verification	src/verify_outputs.py
Results-only master exports	src/create_results_master.py

11. Sources and Reproducibility

src/prepare_data.py (src/prepare_data.py)
src/build_analysis_dataset.py (src/build_analysis_dataset.py)
src/descriptive_analysis.py (src/descriptive_analysis.py)
src/composite_index.py (src/composite_index.py)
src/topsis.py (src/topsis.py)
src/clustering.py (src/clustering.py)
src/sensitivity_analysis.py (src/sensitivity_analysis.py)
src/package_results.py (src/package_results.py)
src/verify_outputs.py (src/verify_outputs.py)
src/create_results_master.py (src/create_results_master.py)

11. Sources and Reproducibility

Exact tested direct dependencies

Package	Version
pandas	2.3.2
xlrd	2.0.2
numpy	2.3.3
scipy	1.16.2
scikit-learn	1.7.2
matplotlib	3.10.6
nbformat	5.10.4
nbclient	0.10.2
ipykernel	7.1.0

11. Sources and Reproducibility

Rebuild these documents only: python src/create_results_master.py

Full precision: results/*.csv. Full scenario scores: results/sensitivity_results.csv. Original calculation settings: docs/methods.md.